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Tsutsumi et al.

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(54) **PRINTING APPARATUS AND PRINthead**

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(52) **U.S. Cl.**
CPC **B41J 2/17523** (2013.01)

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B41J 2/17509; B41J 2/17513; B41J
2/1752; B41J 2/17523; B41J 29/02; B41J
29/13

See application file for complete search history.

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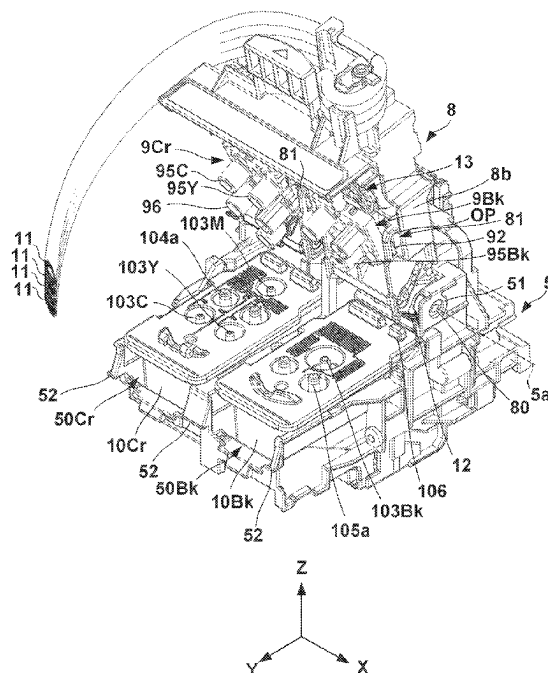
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(57) **ABSTRACT**

A printing apparatus includes a support member configured to support a printhead that performs printing by discharging ink to a print medium, a connecting member including a connecting portion connected to an ink introduction portion of the printhead and through which the ink introduction portion communicates with an ink supply passage via the connecting portion, a rotation member configured to hold the connecting member such that a posture of the connecting member is changeable, and configured to be supported by the support member to be rotatable between a connecting position at which the ink introduction portion and the connecting portion are connected and a connection releasing position at which the ink introduction portion and the connecting portion are not connected, and a biasing portion configured to bias the connecting member to a first posture with respect to the rotation member.

20 Claims, 17 Drawing Sheets



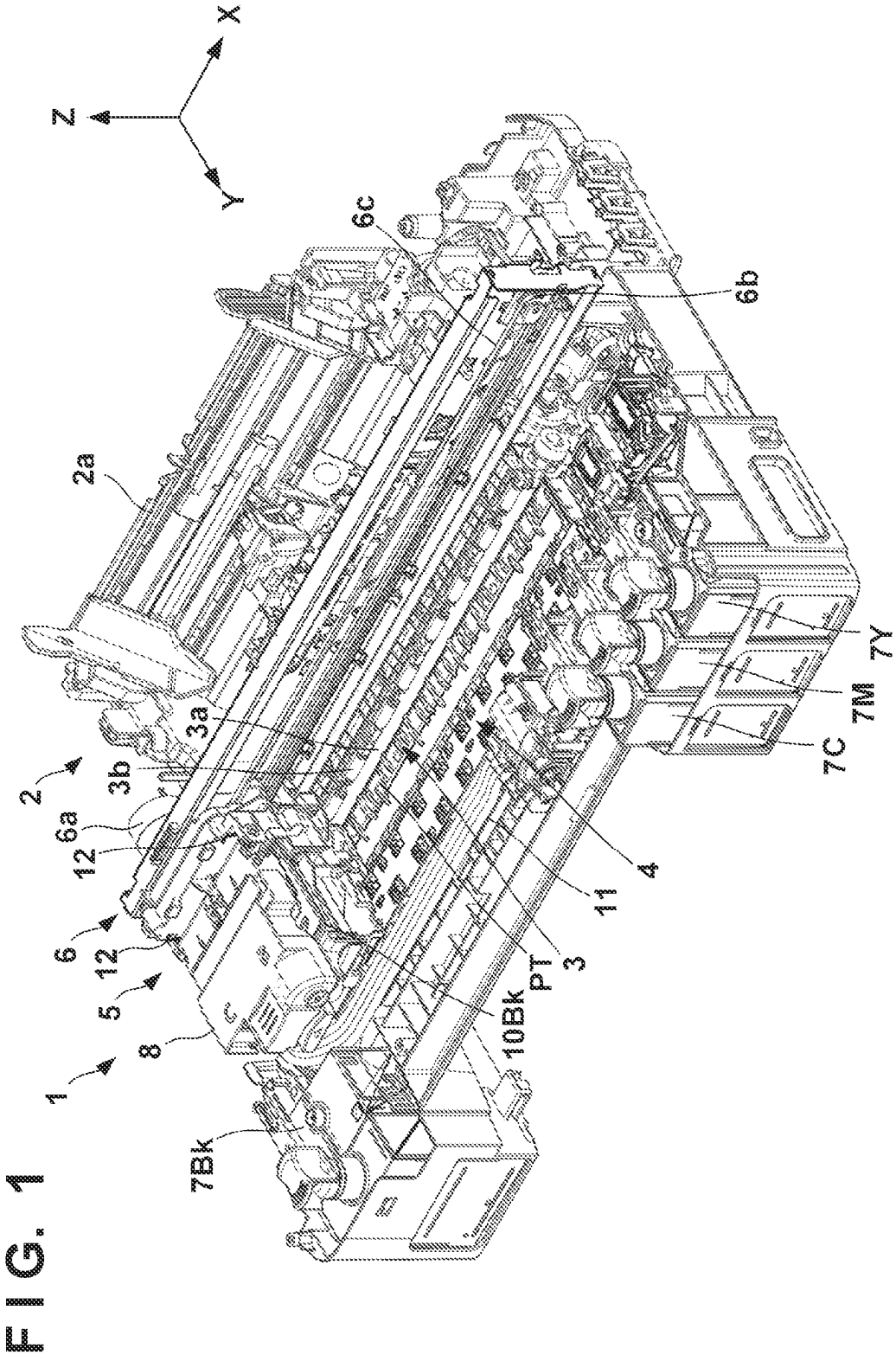


FIG. 2

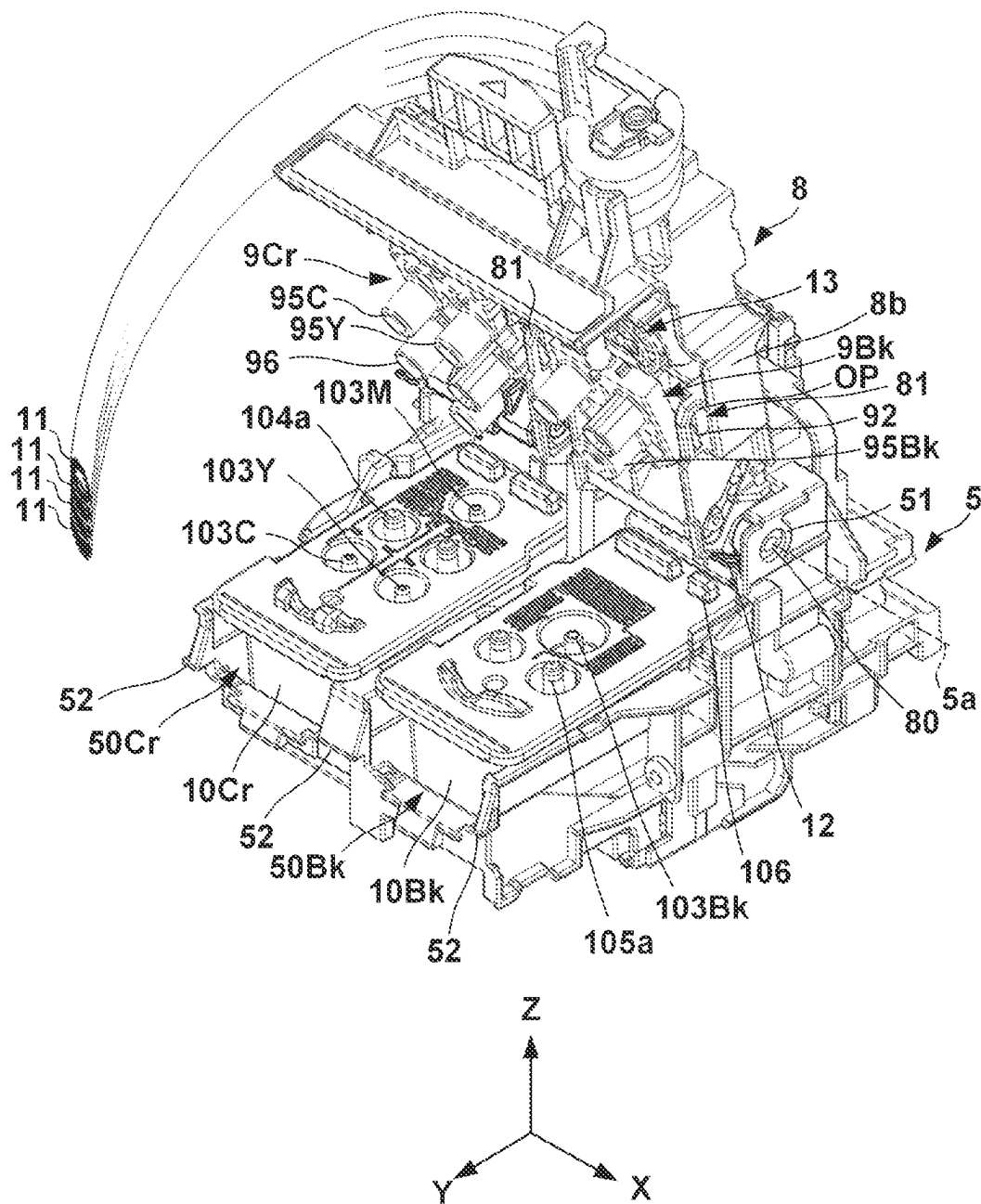


FIG. 3

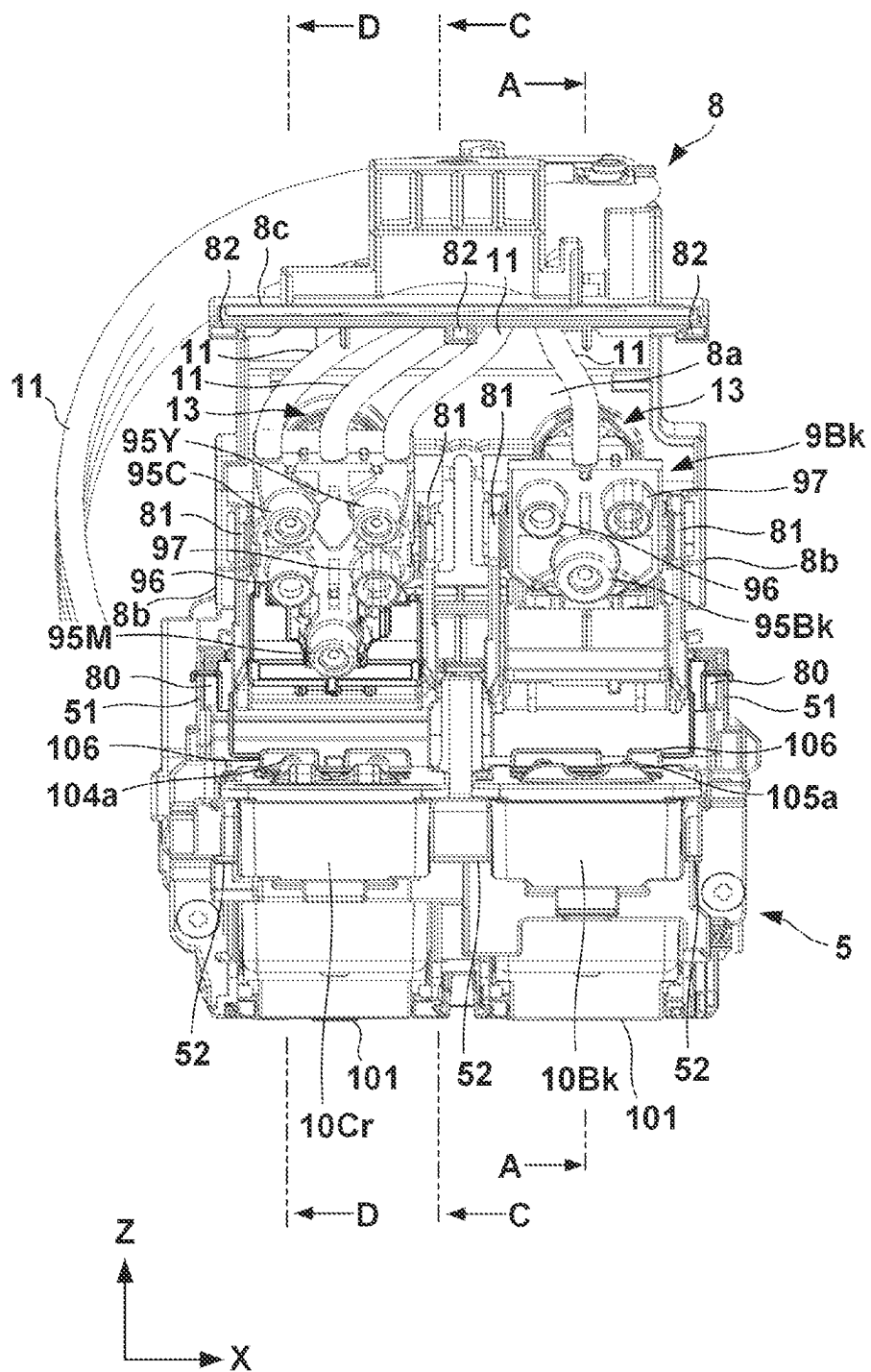
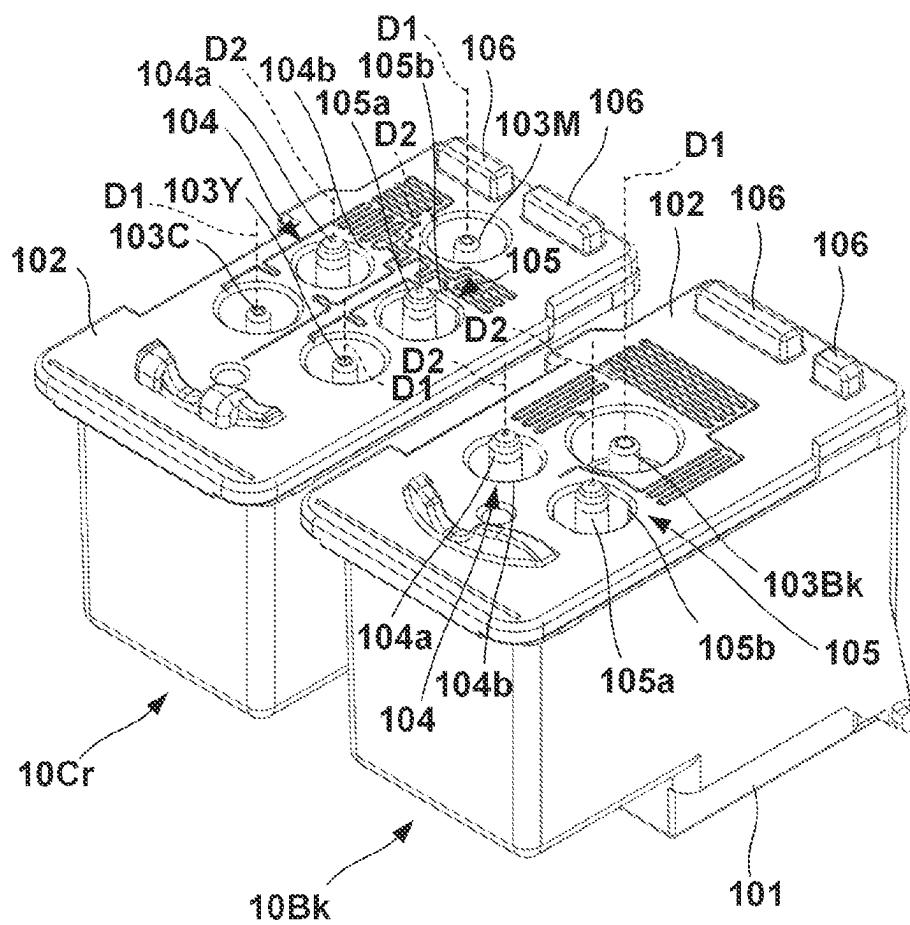
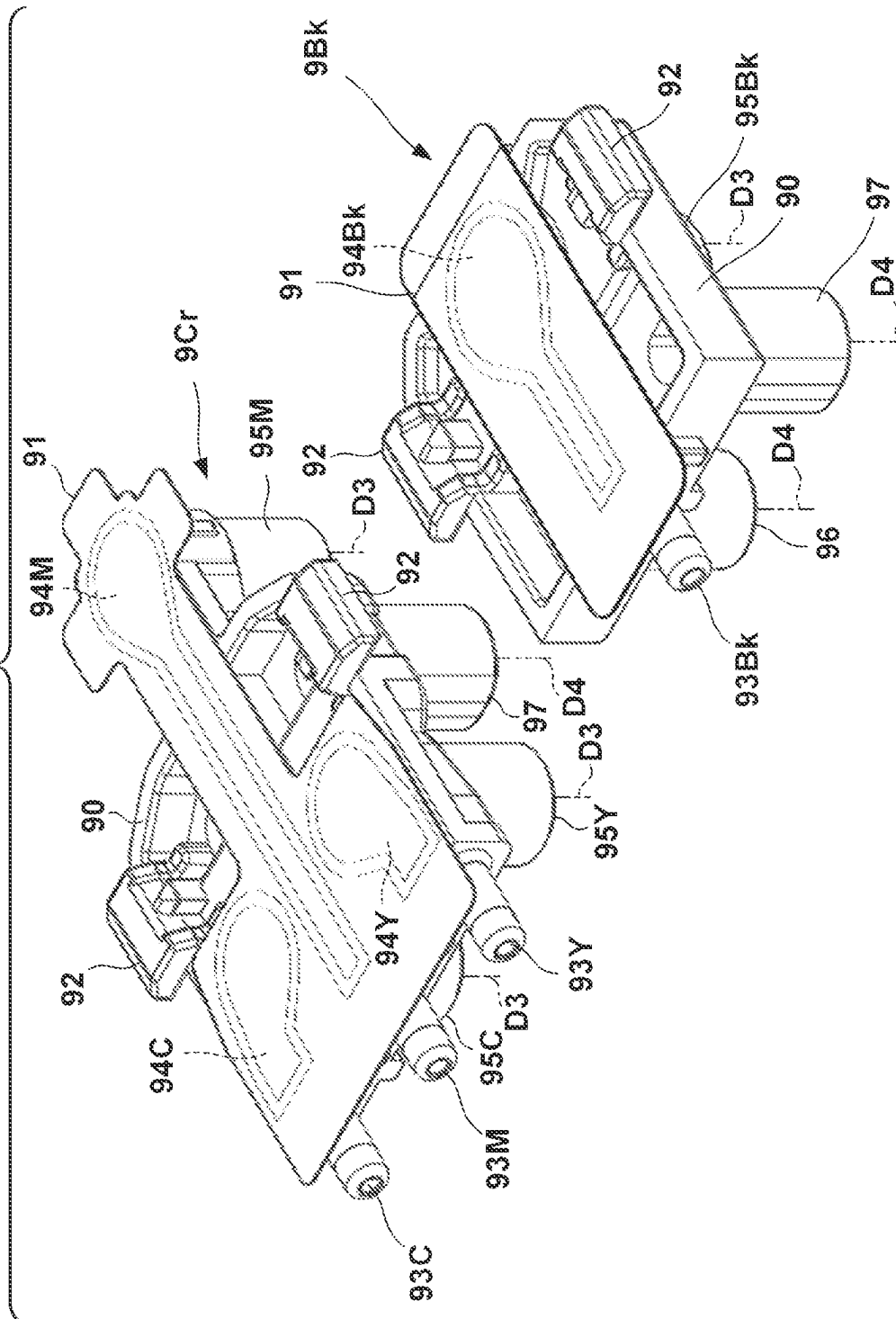


FIG. 4





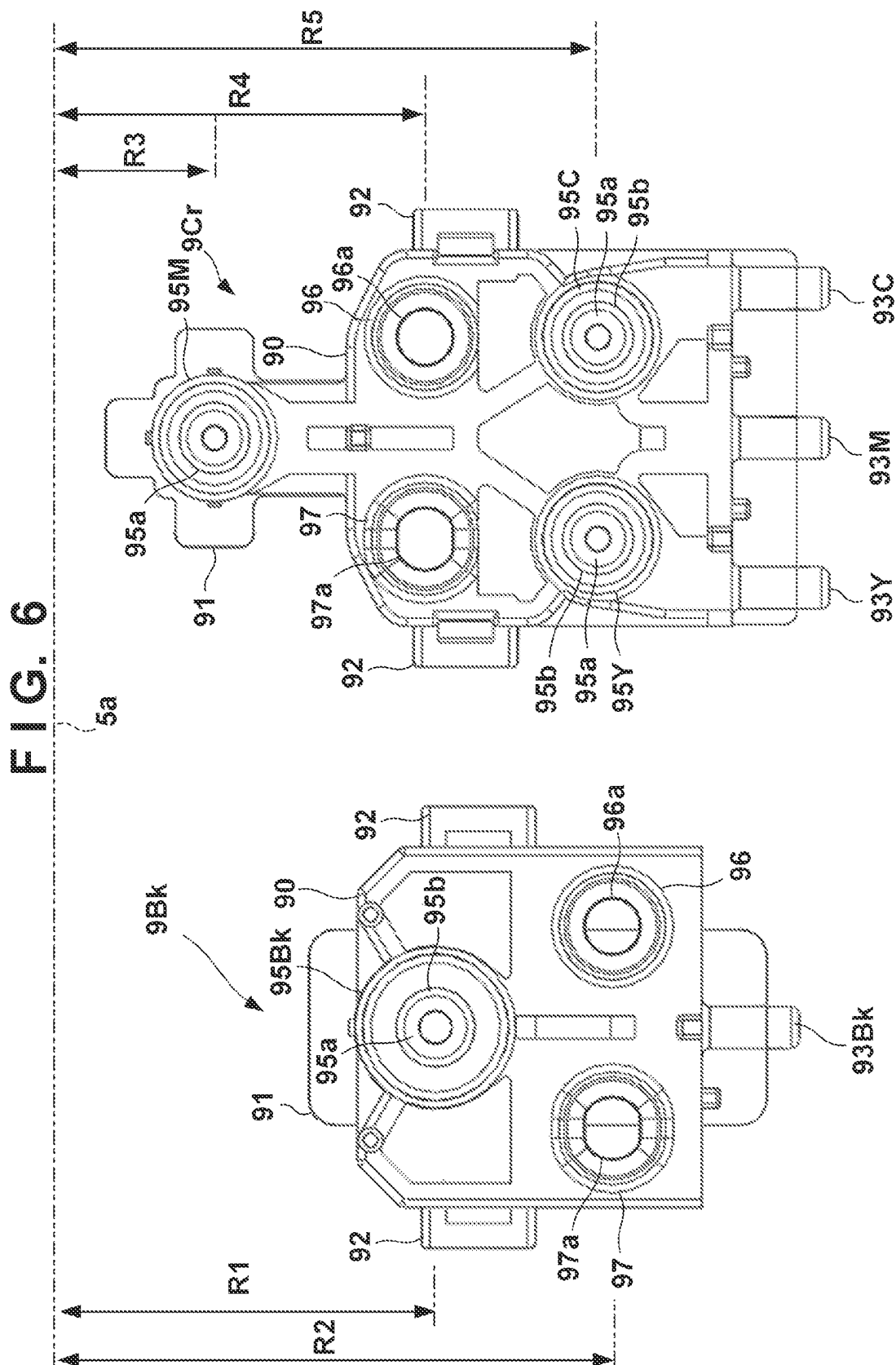
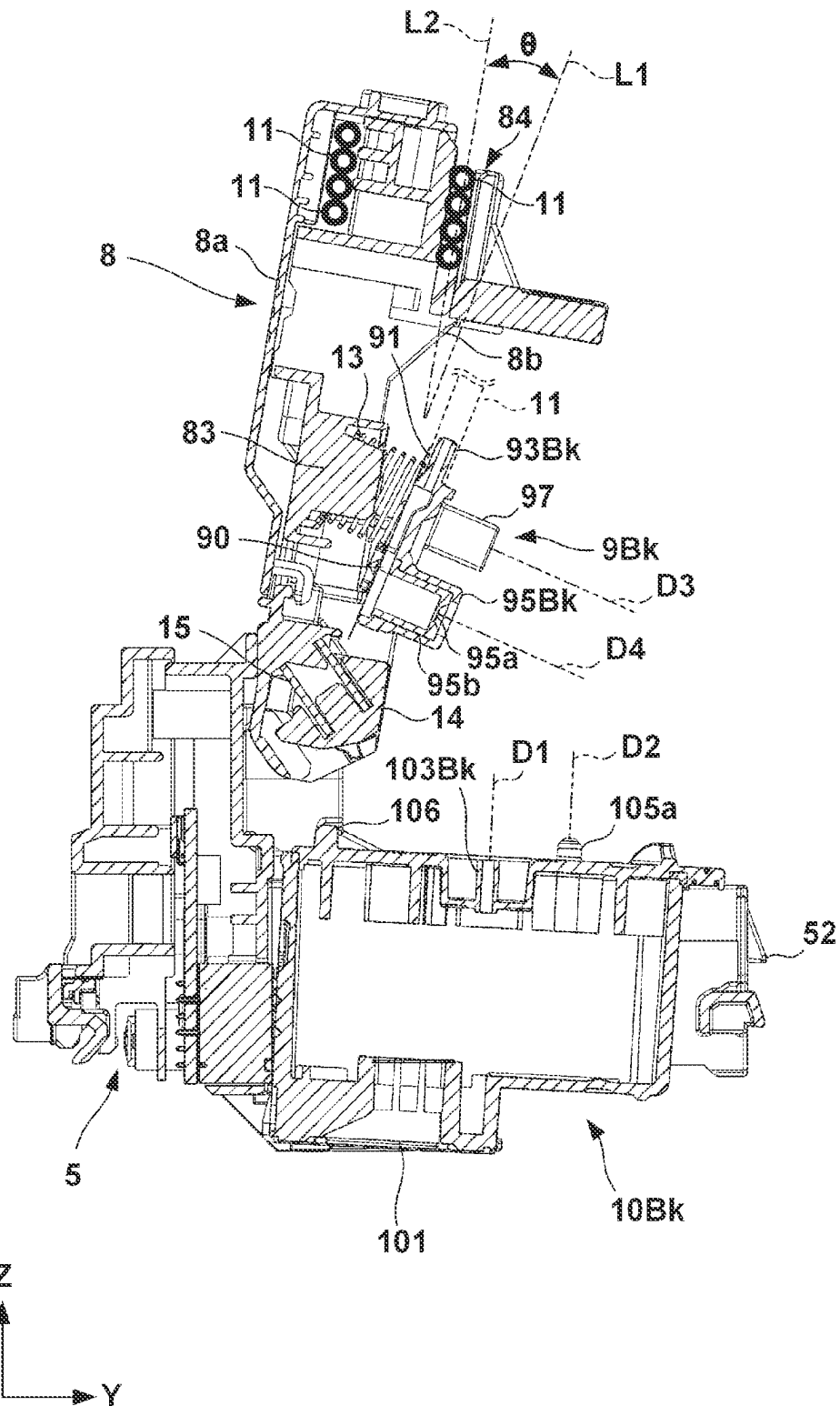


FIG. 7







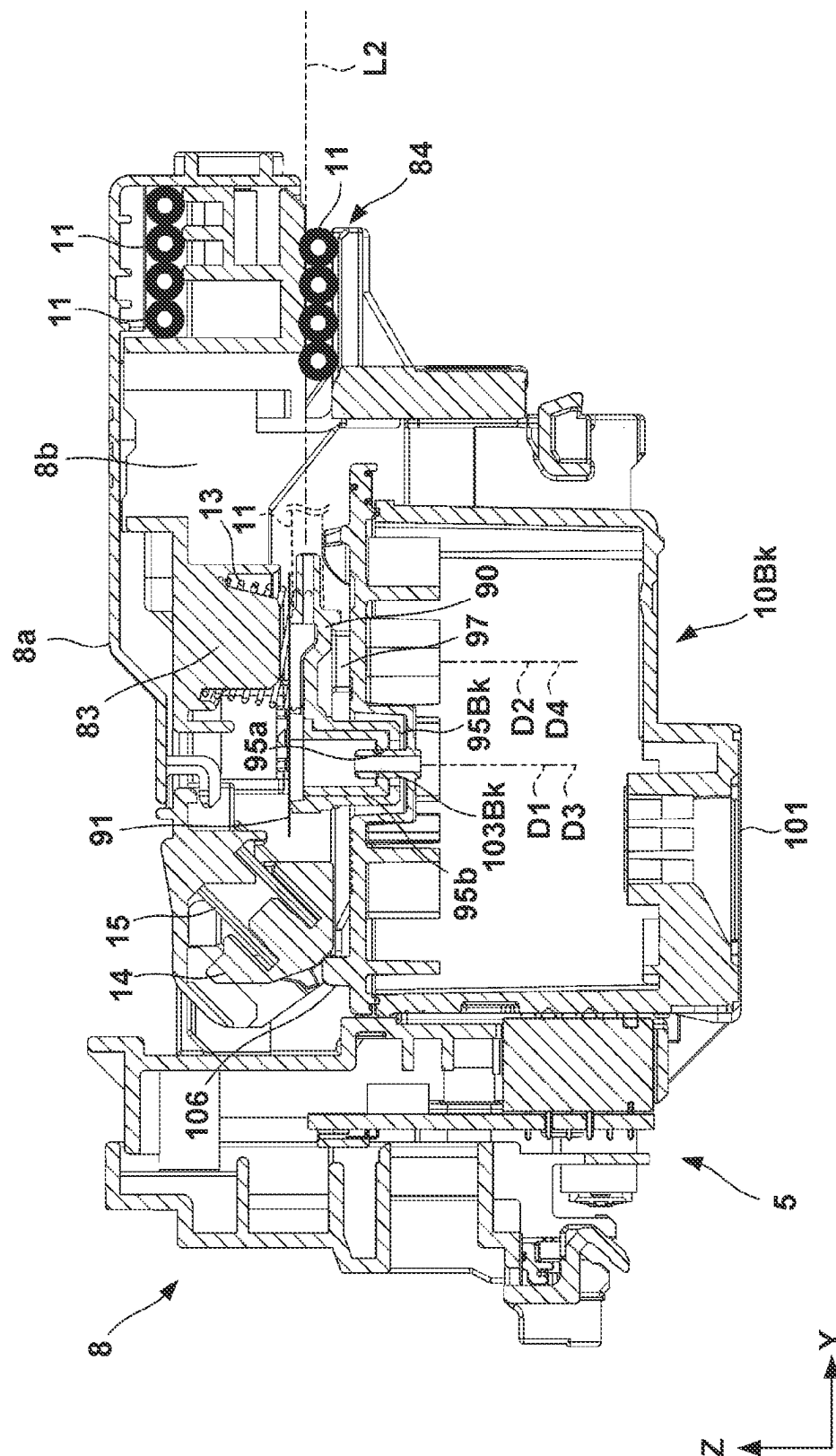



FIG. 9A

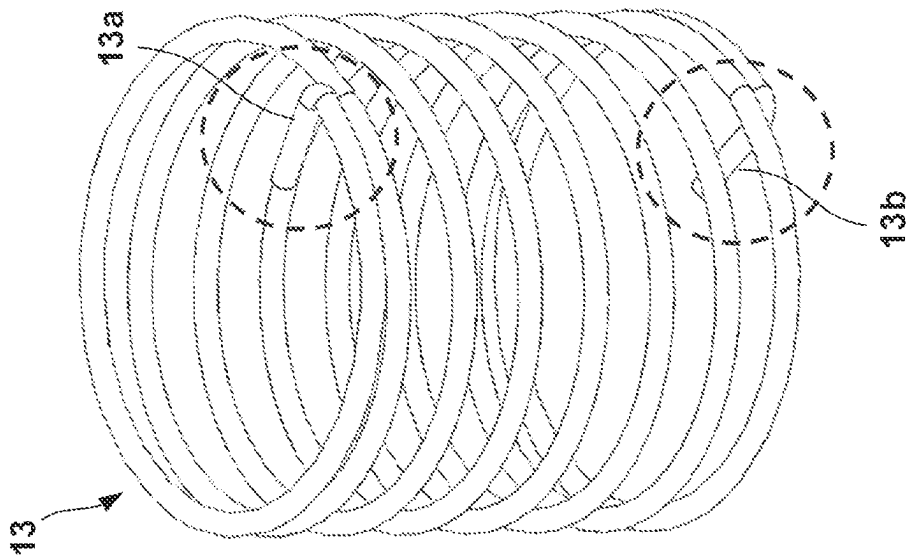


FIG. 9B

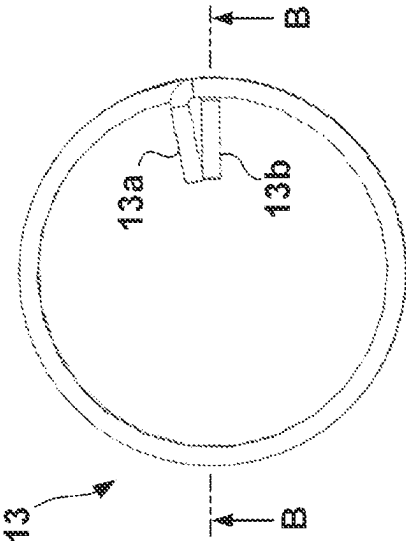


FIG. 9C

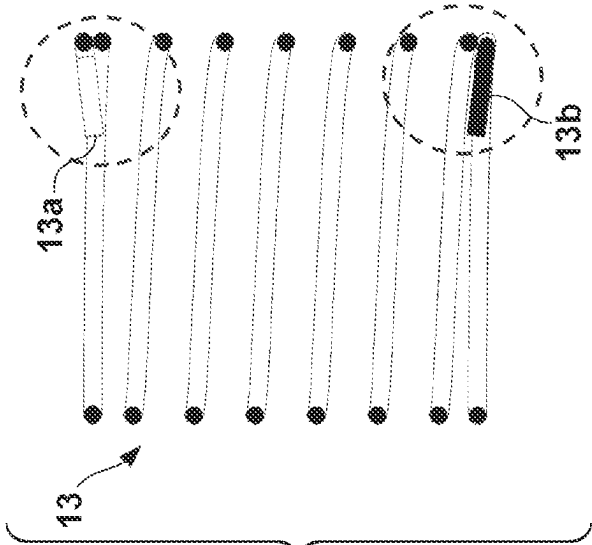


FIG. 10

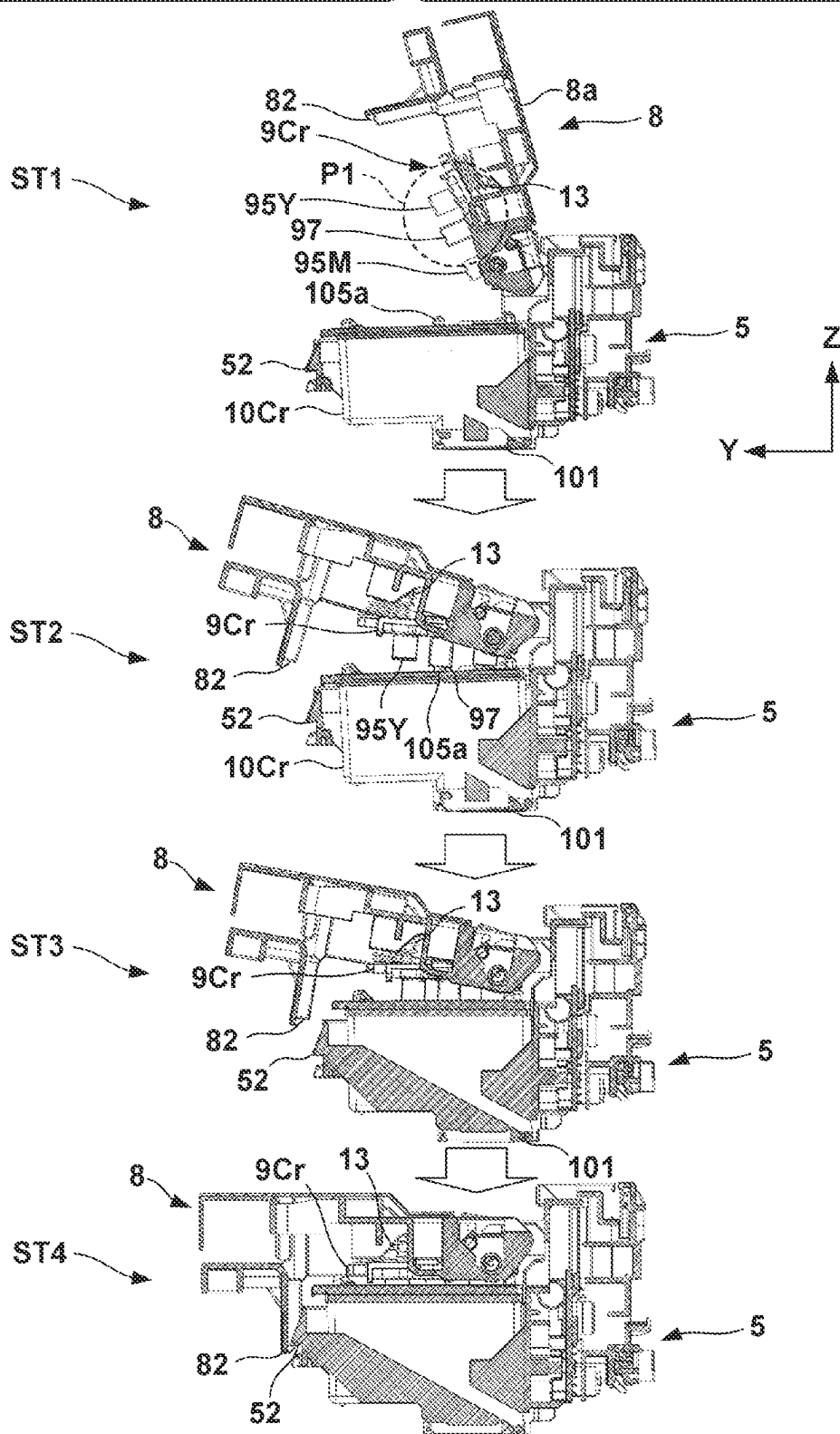


FIG. 11

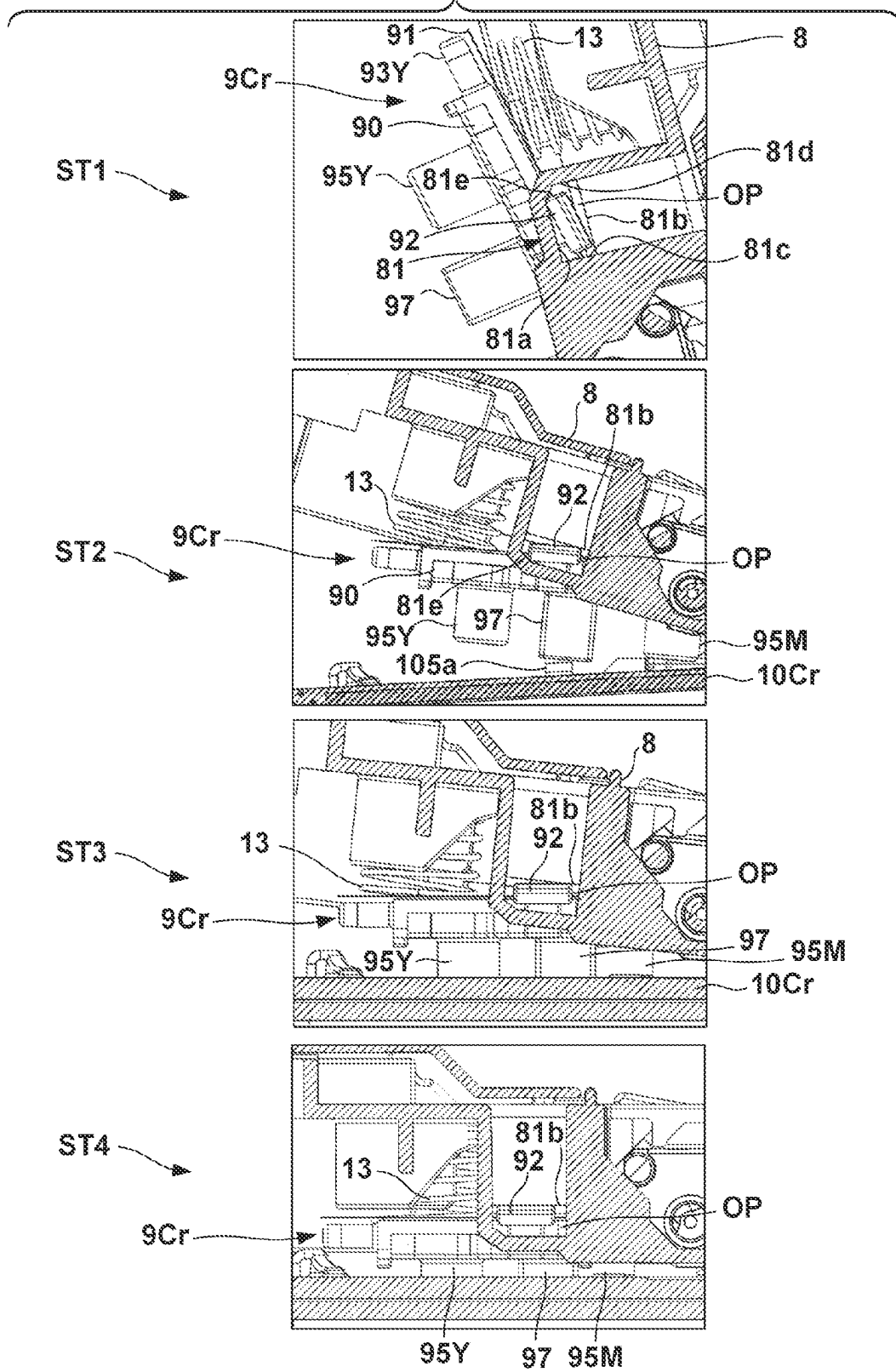


FIG. 12

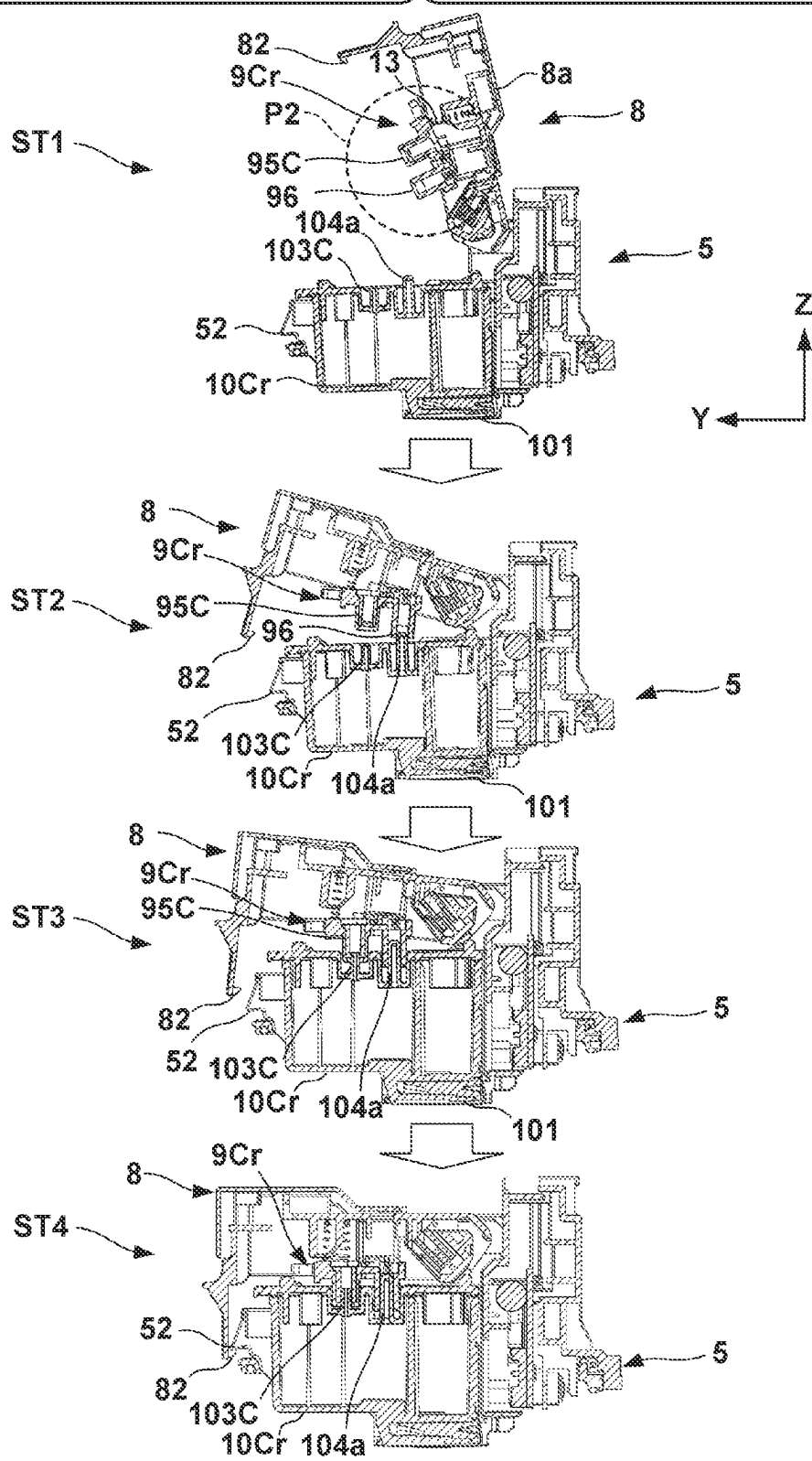


FIG. 13

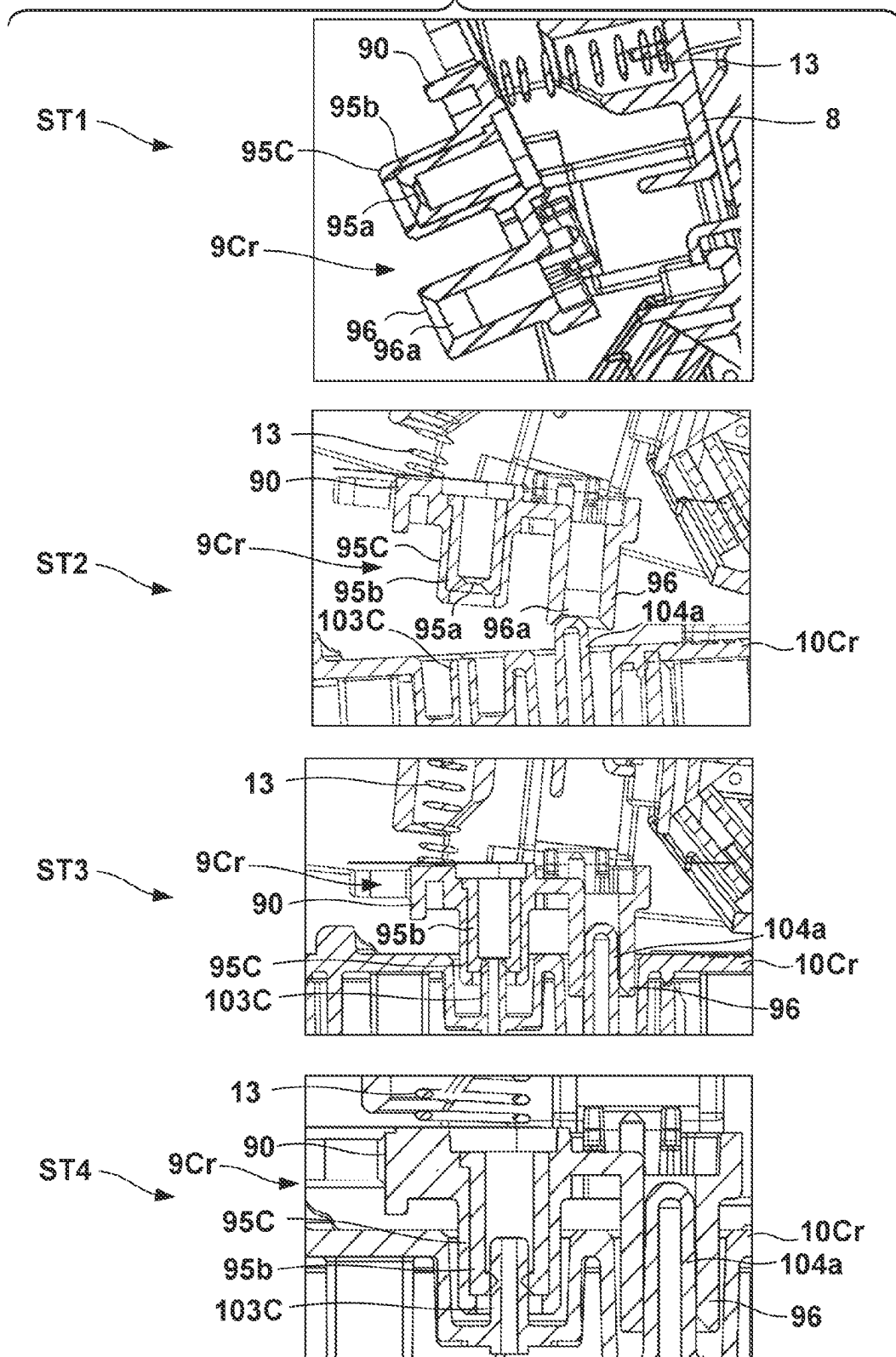


FIG. 14B

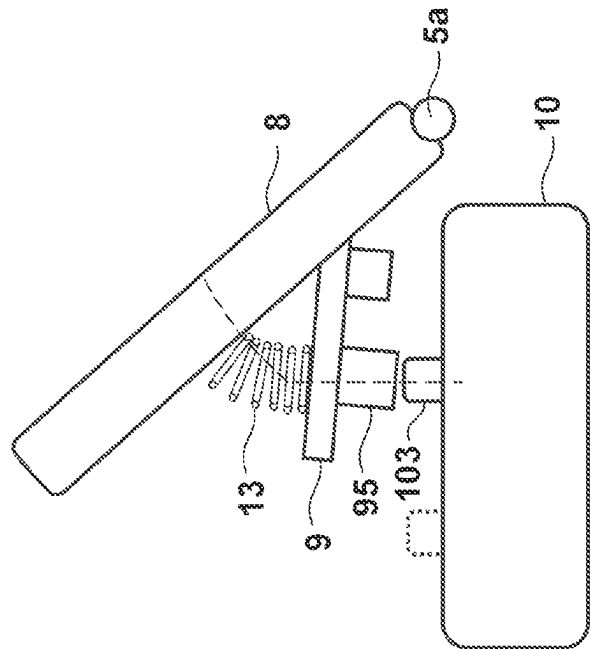


FIG. 14A

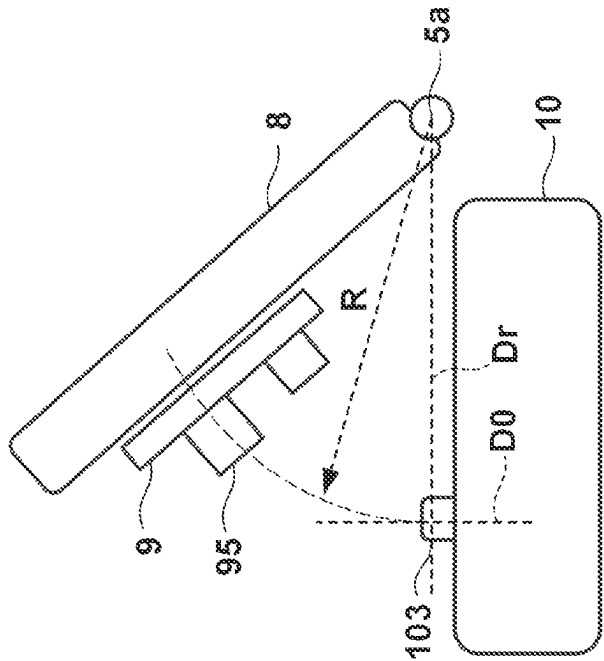


FIG. 15

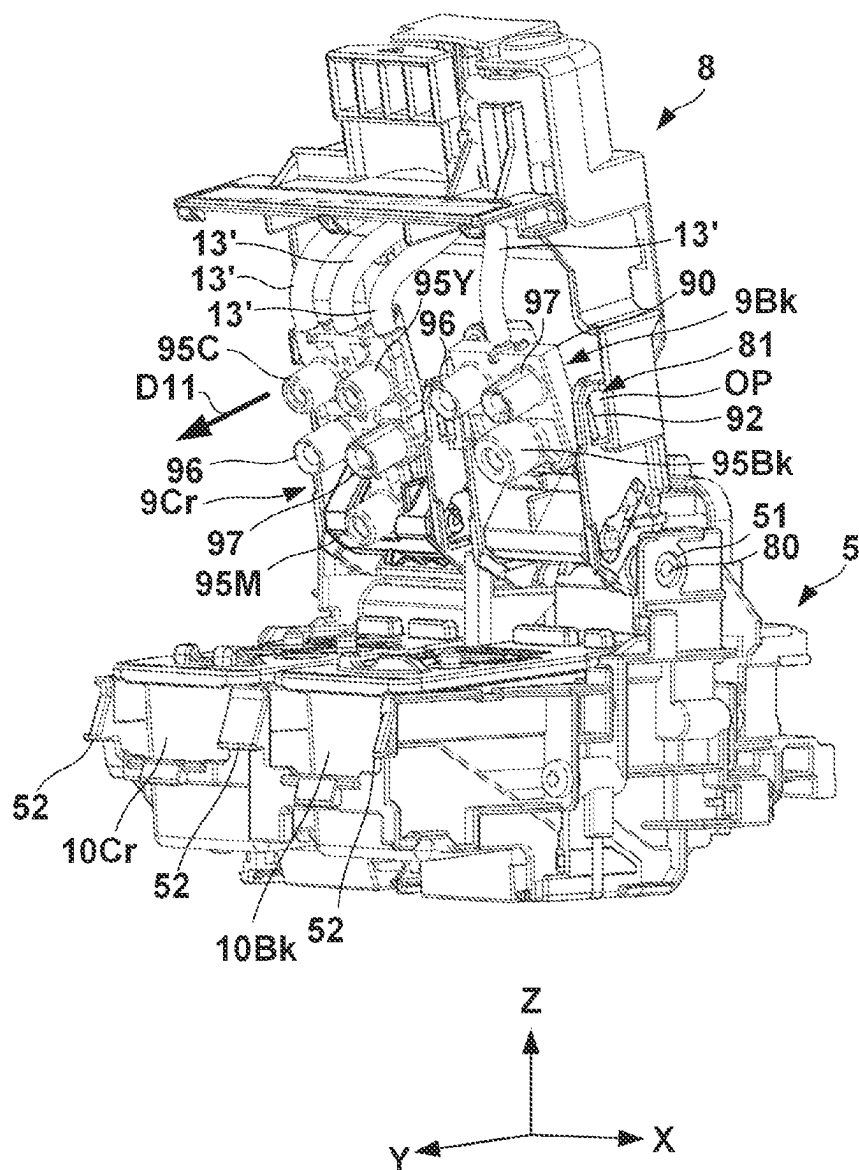


FIG. 16A

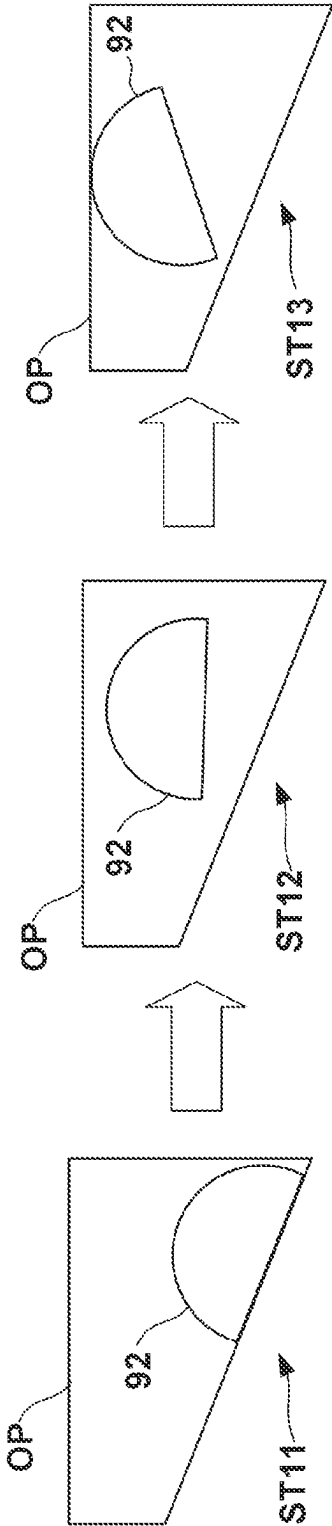


FIG. 16B

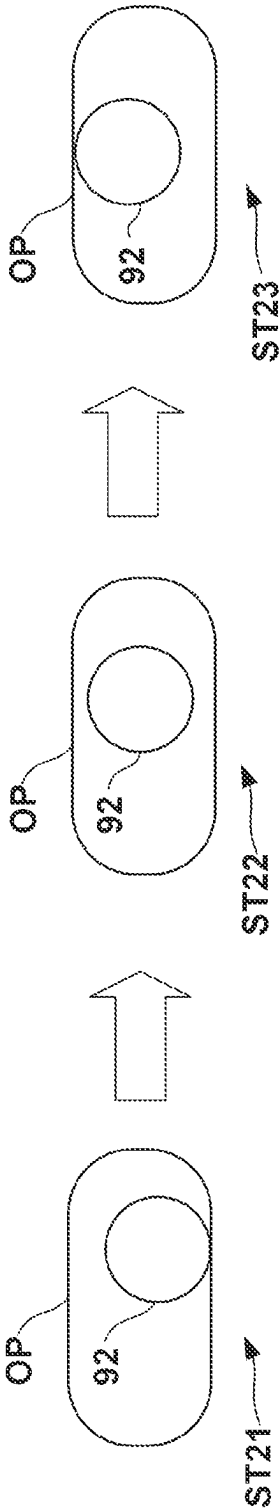
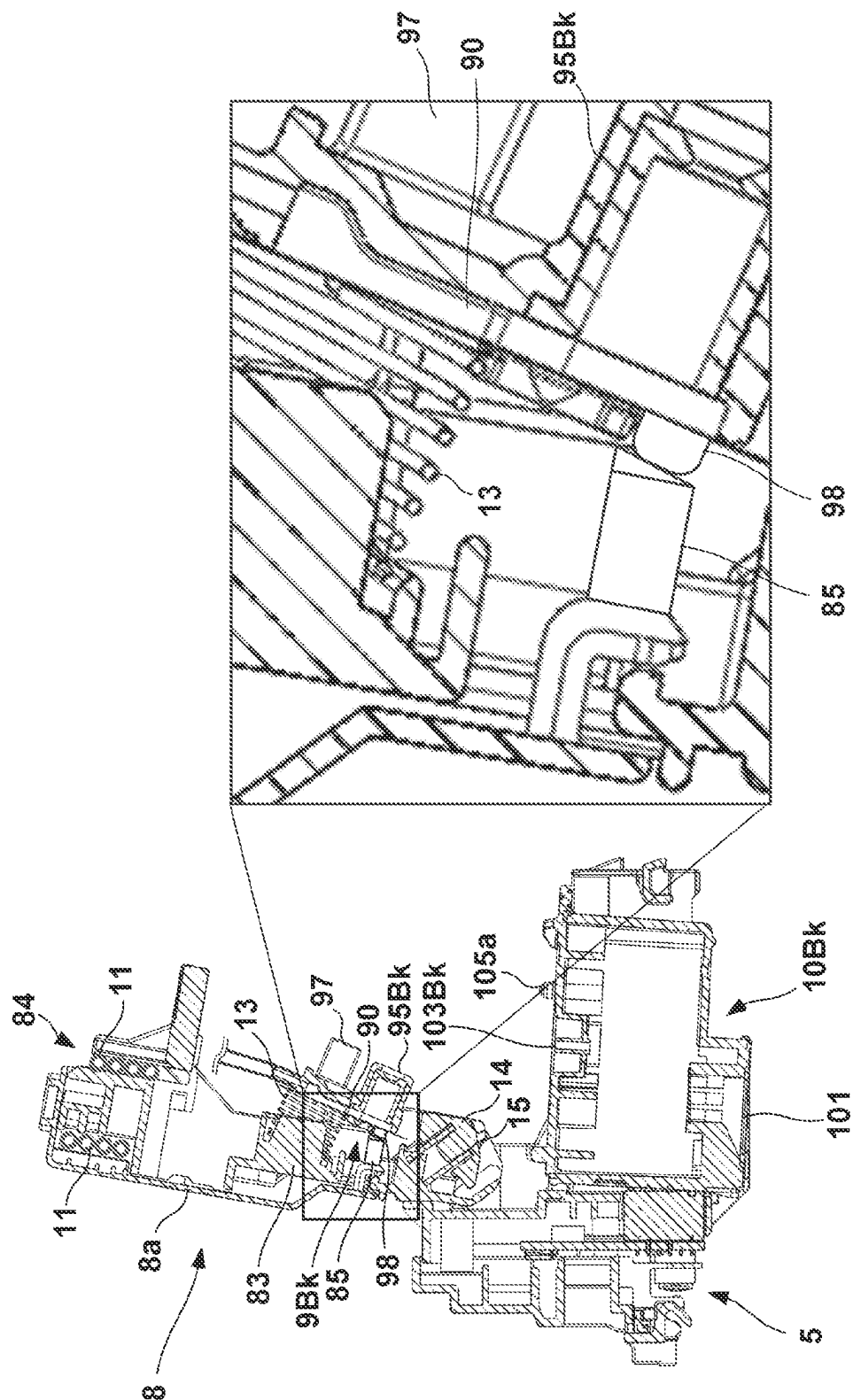


FIG. 17



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PRINTING APPARATUS AND PRINthead**BACKGROUND OF THE INVENTION****Field of the Invention**

The present invention relates to a printing apparatus.

Description of the Related Art

There is proposed a printing apparatus including a member that connects a printhead and an ink supply tube. Japanese Patent No. 3948959 discloses a printing apparatus in which a joint member is rotatably attached to a carriage cover. A tube is connected to the joint member. If the carriage cover is closed, the printhead and the joint member are connected, and ink is supplied from the tube to the printhead. The joint member is loosely fitted in a groove formed in a carriage. When the carriage cover is closed, the joint member is guided by the groove, and its posture with respect to the printhead is controlled.

In the configuration of Japanese Patent No. 3948959, the joint member moves in an arc pattern in accordance with the rotation of the carriage cover, but the groove has a linear shape. If the fitting between the joint member and the groove is tight, it may be difficult for the joint member to smoothly move. If the fitting between the joint member and the groove is loosened, the joint member can smoothly move. However, the posture of the joint member readily changes, and the connection accuracy between the printhead and the joint member may lower due to an unintended change of the posture.

SUMMARY OF THE INVENTION

The present invention provides a technique capable of improving connection accuracy when connecting a connecting member to a printhead.

According to an aspect of the present invention, there is provided a printing apparatus comprising: a support member configured to support a printhead that performs printing by discharging ink to a print medium; a connecting member including a connecting portion connected to an ink introduction portion of the printhead and through which the ink introduction portion communicates with an ink supply passage via the connecting portion; a rotation member configured to hold the connecting member such that a posture of the connecting member is changeable, and configured to be supported by the support member to be rotatable between a connecting position at which the ink introduction portion and the connecting portion are connected and a connection releasing position at which the ink introduction portion and the connecting portion are not connected; and a biasing portion configured to bias the connecting member to a first posture with respect to the rotation member.

Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a view showing the outer appearance of a printing apparatus according to an embodiment of the present invention;

FIG. 2 is a perspective view showing the structure on the periphery of a carriage;

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FIG. 3 is a front view showing the structure on the periphery of the carriage;

FIG. 4 is a perspective view of printheads;

FIG. 5 is a perspective view of connecting members;

FIG. 6 is a bottom view of the connecting members;

FIG. 7 is a sectional view taken along a line A-A in FIG. 3;

FIG. 8 is a sectional view showing a state in which ahead set cover is located at a closed position;

FIGS. 9A to 9C are explanatory views of a biasing portion;

FIG. 10 is a view for explaining the operations of the head set cover and the connecting member;

FIG. 11 is a view for explaining the operations of the head set cover and the connecting member;

FIG. 12 is a view for explaining the operations of the head set cover and the connecting member;

FIG. 13 is a view for explaining the operations of the head set cover and the connecting member;

FIGS. 14A and 14B are explanatory views of a principle;

FIG. 15 is a view showing a mode in which a tube is used as a biasing portion;

FIGS. 16A and 16B are views showing another example of an opening portion and a claw portion; and

FIG. 17 is a view showing another example of a configuration for regulating the posture of the connecting member.

DESCRIPTION OF THE EMBODIMENTS

Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar elements or configurations, and redundant description thereof is omitted.

First Embodiment**"1. Outline of Printing Apparatus"**

FIG. 1 is a perspective view of a printing apparatus 1 according to an embodiment of the present invention. The printing apparatus 1 according to this embodiment is an inkjet printing apparatus that performs printing on a print medium by discharging ink. However, the present invention can also be applied to various kinds of printing apparatuses other than the inkjet printing apparatus. In the drawings, arrows X and Y indicate directions crossing each other, and these are orthogonal to each other in this embodiment. An arrow Z indicates a vertical direction (gravity direction). The X direction is the widthwise direction of the printing apparatus 1 (the left-and-right direction, or the widthwise direction of a print medium). The Y direction is the depth direction (front-and-rear direction) of the printing apparatus 1.

Note that "printing" includes not only forming significant information such as characters and graphics but also forming images, figures, patterns, and the like on print media in a broad sense, or processing print media, regardless of whether the information formed is significant or insignificant or whether the information formed is visualized so that a human can visually perceive it. In addition, although in this embodiment, sheet-like paper is assumed as a "print medium", cloth, a plastic film, and the like may also be used.

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The printing apparatus 1 includes a feed unit 2, a conveyance unit 3, and a discharge unit 4, which are configured to convey a print medium. The feed unit 2 includes a tray 2a on which sheet-like print media are stacked, and a feed mechanism (not shown) for the print medium. The feed mechanism includes, for example, a feed roller that feeds the print medium on the tray 2a, and a feed motor that is a driving source for rotating the feed roller.

The conveyance unit 3 is a mechanism that conveys, in the Y direction (sub-scanning direction), the print medium fed from the feed unit 2. The conveyance unit 3 includes a conveyance roller 3a, and a conveyance motor (not shown) that is a driving source for rotating the conveyance roller 3a. Pinch rollers 3b are pressed against the conveyance roller 3a, and the print medium is clamped in the nip portions between these. When the conveyance roller 3a rotates, the print medium is conveyed on a platen PT.

The discharge unit 4 is a mechanism that further conveys, in the Y direction (sub-scanning direction), the print medium conveyed from the conveyance unit 3. The discharge unit 4 includes a discharge roller, a conveyance motor that is a driving source for rotating the discharge roller, and spurs in contact with the discharge roller (none are shown), and discharges the print medium to the front side of the printing apparatus 1.

Containers 7Bk, 7C, 7M, and 7Y (to be referred to as containers 7 hereinafter generically or without distinction) are ink tanks in which liquid inks are stored. In this embodiment, the container 7 is a stationary type container fixed in the printing apparatus 1. If the remaining ink amount decreases, a user replenishes ink to the container 7 without detaching the container 7 from the printing apparatus 1. Inks of different types are stored in the four containers 7. In this embodiment, black ink is stored in the container 7Bk, cyan ink is stored in the container 7C, magenta ink is stored in the container 7M, and yellow ink is stored in the container 7Y. Note that the types of ink are not limited to four types, as in this embodiment, and one type of ink may be used, or a plurality of types other than the four types may be used. The number of containers 7 need only be equal to or more than the number of types of inks.

The printing apparatus 1 includes a carriage 5. The carriage 5 is a support member that supports a printhead 10Cr (not shown in FIG. 1) and a printhead 10Bk (to be referred to as printheads 10 hereinafter generically or without distinction). The carriage 5 is covered with a head set cover 8 from above. The printhead 10 according to this embodiment is a printhead that performs printing by discharging ink supplied from the container 7 to a print medium. The printhead 10 includes a discharge surface 101 (see FIGS. 3 and 4) with a plurality of nozzles for discharging ink. The discharge surface 101 is arranged to face the platen PT. Each nozzle is provided with, for example, an electrothermal transducer (heater). When the electrothermal transducer is energized, ink is heated and foamed. The ink is discharged by the foaming energy. A structure that discharges ink using a piezoelectric element in place of the electrothermal transducer may be used.

The carriage 5 is reciprocally moved in the X direction (main scanning direction) by the driving unit 6. The driving unit 6 includes a driving pulley and a driven pulley (only a driven pulley 6b is shown in FIG. 1) which are arranged apart in the X direction, an endless belt 6c wound around the pulleys, and a carriage motor 6a that is a driving source for rotating the driving pulley. The carriage 5 is connected to the endless belt 6c. When the endless belt 6c is made to travel, the carriage 5 moves in the X direction.

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In the process of the movement of the carriage 5, ink is discharged from the printhead 10 to the print medium on the platen PT, thereby printing an image. This operation is sometimes called print scanning. A printing operation is performed by alternately repeating a print medium conveyance operation by the conveyance unit 3 and print scanning. That is, the conveyance unit 3 intermittently conveys the print medium, and print scanning is performed when the print medium conveyance is stopped, thereby printing an image on the print medium.

As described above, the printing apparatus 1 according to this embodiment is a serial type inkjet printing apparatus in which the printhead 10 is mounted on the carriage 5 that reciprocally moves in the X direction. However, the present invention can also be applied to another printing apparatus such as an inkjet printing apparatus including a so-called full-line head (printhead) provided with a plurality of nozzles configured to discharge a liquid to a region corresponding to the width of a print medium.

"2. Structure on Periphery of Carriage"

The structure on the periphery of the carriage 5 will be described with reference to FIGS. 2 to 9 in addition to FIG. 1. FIGS. 2 and 3 will mainly be referred to. FIG. 2 is a perspective view showing the structure on the periphery of the carriage 5, and FIG. 3 is a front view showing the structure on the periphery of the carriage 5. FIGS. 2 and 3 show a state in which the head set cover 8 is located at the open position.

<Printheads>

In this embodiment, a plurality of printheads 10 are supported by the carriage 5. More specifically, two types of printheads 10 including the printhead 10Cr and the printhead 10Bk are supported. The printhead 10Cr is a printhead that discharges cyan ink, magenta ink, and yellow ink. The printhead 10Bk is a printhead that discharges black ink. In this embodiment, the printheads 10 are exchangeably supported by the carriage 5. Since the printheads 10 are exchangeable, formally, the printheads 10 and the apparatus main body other than those are called the printing apparatus 1, but the apparatus main body is called the printing apparatus 1 in some cases.

FIG. 4 will also be referred to. FIG. 4 is a perspective view of the printheads 10Cr and 10Bk. The printhead 10 has a rectangular parallelepiped shape as a whole, and includes the discharge surface 101 on the bottom portion. An upper surface 102 of the printhead 10 is provided with the following components.

The upper surface 102 of the printhead 10Bk is provided with an engaging portion 106 projecting upward from the upper surface 102, an ink introduction portion 103Bk, and posture control portions 104 and 105. The engaging portion 106 is an engaging portion for positioning when the printhead 10Bk is attached to the carriage 5. The ink introduction portion 103Bk is a portion serving as an inlet to introduce black ink stored in the container 7Bk to the printhead 10Bk, and is a tube member extending in the vertical direction. A circular concave portion is formed around the ink introduction portion 103Bk. The posture control portions 104 and 105 are members that change the posture of a connecting member 9Bk. The posture control portion 104 includes a shaft portion 104a having a columnar shape (circular section), and a concave portion 104b having a cylindrical shape. The diameter of the distal end of the shaft portion 104a is gradually reduced. The posture control portion 105 includes a shaft portion 105a having an elliptic columnar shape (elliptic section), and a concave portion 105b having an

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elliptic cylindrical shape. The diameter of the distal end of the shaft portion **105a** is gradually reduced.

The upper surface **102** of the printhead **10Cr** is provided with the engaging portion **106** projecting upward from the upper surface **102**, ink introduction portions **103Y**, **103M**, and **103C**, and the posture control portions **104** and **105**. The engaging portion **106** is an engaging portion for positioning when the printhead **10Cr** is attached to the carriage **5**. The ink introduction portions **103Y**, **103M**, and **103C** are portions serving as inlets to introduce yellow ink, magenta ink, and cyan ink stored in the containers **7Y**, **7M**, and **7C** to the printhead **10Cr**, and are tube members extending in the vertical direction. A circular concave portion is formed around each of the ink introduction portions **103Y**, **103M**, and **103C**. The posture control portions **104** and **105** are members that change the posture of a connecting member **9Cr**. The posture control portion **104** includes the shaft portion **104a** having a columnar shape, and the concave portion **104b** having a cylindrical shape. The diameter of the distal end of the shaft portion **104a** is gradually reduced. The posture control portion **105** includes the shaft portion **105a** having an elliptic columnar shape, and the concave portion **105b** having an elliptic cylindrical shape. The diameter of the distal end of the shaft portion **105a** is gradually reduced.

Referring to FIG. 4, arrows **D1** indicate the extending directions (axial directions) of the ink introduction portions **103Bk**, **103Y**, **103M**, and **103C** (to be referred to as ink introduction portions **103** hereinafter generically or without distinction). The arrows **D1** also indicate the connecting directions between the ink introduction portions **103Bk**, **103Y**, **103M**, and **103C** and connecting portions **95Bk**, **95Y**, **95M**, and **95C** (to be described later). Arrows **D2** indicate the extending directions (axial directions) of the shaft portions **104a** of the posture control portions **104** and **105**, and also indicate the connecting directions between the posture control portions **104** and **105** and engaging portions **96** and **97** (to be described later). In this embodiment, the directions **D1** and the directions **D2** are parallel.

<Carriage and Head Set Cover>

FIGS. 1 to 3 will mainly be referred to. The carriage **5** includes a head storage portion **50Cr** that stores the printhead **10Cr**, and a head storage portion **50Bk** that stores the printhead **10Bk**. The head storage portion **50Cr** and the head storage portion **50Bk** are arranged in the X direction. For this reason, the printhead **10Cr** and the printhead **10Bk** are arranged in the X direction, too. The carriage **5** includes an opening portion in the bottom portion, and the discharge surfaces **101** of the printhead **10Cr** and the printhead **10Bk** are exposed to the outside (lower side) of the carriage **5**. On the front portion of the carriage **5**, a plurality of lock portions **52** that maintain the head set cover **8** at the closed position are provided apart in the X direction.

The head set cover **8** is a rotation member that is rotatably supported by the carriage **5** and enables exchange of the printhead **10**. The head set cover **8** includes an upper wall portion **8a**, left and right side wall portions **8b**, and a front wall portion **8c**, and has a box shape open to the lower and rear sides as a whole.

The carriage **5** is provided with rotation shafts **51** apart to the left and right sides. The left and right side wall portions **8b** of the head set cover **8** are provided with bearings **80** that support the rotation shafts **51**, and the head set cover **8** can rotate, with respect to the carriage **5**, about a rotation center line **5a** that is the center of the rotation shafts **51**. In this embodiment, the direction of the rotation center line **5a** is the X direction.

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An elastic member **12** that biases the head set cover **8** to the open position is provided near each rotation shaft **51**. In this embodiment, the elastic member **12** is a torsion coil spring through which the rotation shaft **51** is inserted. A plurality of lock portions **82** corresponding to the lock portions **52** of the carriage **5** are provided on the front wall portion **8c** of the head set cover **8**. FIG. 7 is a sectional view taken along a line A-A in FIG. 3, and shows a state in which the head set cover **8** is located at the open position. FIG. 8 is a sectional view corresponding to the line A-A in FIG. 3, and shows a state in which the head set cover **8** is located at the closed position. The lock portions **52** and the lock portions **82** each have a wedge shape, and the head set cover **8** is maintained at the closed position by locking the lock portions **82** on the lock portions **52**. When the lock between the lock portions **52** and the lock portions **82** is released, the head set cover **8** is automatically rotated to the open position by the biasing of the elastic members **12**.

<Connecting Members>

FIGS. 1 to 3, 7, and 8 will mainly be referred to. The connecting members **9Cr** and **9Bk** (to be referred to as connecting members **9** hereinafter generically or without distinction) are held by the head set cover **8**. The connecting members **9** are arranged inside the upper wall portion **8a**. If the head set cover **8** is located at the closed position (FIG. 8), the upper wall portion **8a** covers the printheads **10** from above. In this respect, the upper wall portion **8a** can be called a cover portion. If the head set cover **8** is located at the closed position (FIG. 8), the connecting members **9** are located between the upper wall portion **8a** and the printheads **10**.

The connecting members **9** are held by a plurality of holding portions **81** of the head set cover **8** such that the postures of the connecting members **9** can be changed. Each holding portion **81** is formed to project inward from the upper wall portion **8a**, and each connecting member **9** is held by a pair of holding portions **81** apart to the left and right sides. In this embodiment, to hold the two connecting members **9**, a total of four holding portions **81** are provided on the head set cover **8**.

The connecting members **9** are members that relay the printheads **10** and tubes **11**. A total of four tubes **11** are provided in correspondence with the types of ink, and each tube **11** is connected to a corresponding one of the containers **7Bk**, **7C**, **7M**, and **7Y**. Each tube **11** is a long member with flexibility and forms an ink supply passage. The four tubes are held and fixed at their middle portions by a groove-shaped holding portion **84** provided in the head set cover **8**.

FIGS. 5 and 6 will be referred to together. FIG. 5 is a perspective view of the connecting members **9Cr** and **9Bk**. FIG. 6 is a bottom view of the connecting members **9Cr** and **9Bk**.

The connecting member **9Bk** includes a main body **90**, and a sealing member **91**. The main body **90** integrally includes a flow-in portion **93Bk** provided at the front end portion, a pair of claw portions **92** provided on the left and right side portions, and the connecting portion **95Bk** and the engaging portions **96** and **97** which are provided on the bottom surface. The tube **11** that forms the ink supply passage for black ink is connected to the flow-in portion **93Bk**. The connecting portion **95Bk** is a member connected to the ink introduction portion **103Bk** of the printhead **10Bk**. The connecting portion **95Bk** according to this embodiment is a tubular member, and a connecting port **95a** to/from which the ink introduction portion **103Bk** is inserted/removed is formed in it. An elastic member **95b** is provided on the inner circumferential surface of the connecting portion

95Bk (FIGS. 7 and 8). The elastic member **95b** is a sealing elastomer that seals the connecting port **95a** and the ink introduction portion **103Bk** and is, for example, molded integrally with the connecting portion **95Bk**.

A groove-shaped relay passage **94Bk** through which the flow-in portion **93Bk** and the connecting portion **95Bk** communicate is formed on the upper surface of the main body **90**. The sealing member **91** is, for example, a sheet-shaped film welded to the upper surface of the main body **90**, and the relay passage **94Bk** is sealed by the sealing member **91**. When the connecting portion **95Bk** and the ink introduction portion **103Bk** are connected, the ink supply passage of the tube **11** and the ink introduction portion **103Bk** communicate, and black ink is supplied from the container **7Bk** to the printhead **10Bk** via the tube **11** and the relay passage **94Bk**.

The pair of claw portions **92** are plate-shaped members and project from the main body **90** in the left-and-right direction (the direction of the rotation axial line **5a**) in a plan view of the connecting member **9Bk**. Each holding portion **81** includes an opening portion **OP** (FIG. 2). The area of the opening portion **OP** is larger than the sectional area of the claw portion **92**. When the claw portions **92** are inserted into the opening portions **OP**, the connecting member **9Bk** is held by the head set cover **8** such that its posture can be changed. The engaging portions **96** and **97** correspond to the posture control portions **104** and **105** of the printhead **10Bk**, respectively. The engaging portion **96** is arranged to engage with the posture control portion **104**, and the engaging portion **97** is arranged to engage with the posture control portion **105**. The engaging portions **96** and **97** are tubular members with holes **96a** and **97a**. The engaging portion **96** has a cylindrical shape in correspondence with the shape of the posture control portion **104**, and the engaging portion **97** has an elliptic cylindrical shape in correspondence with the shape of the posture control portion **105**.

As shown in FIG. 6, a distance **R1** from the rotation center line **5a** to the connecting portion **95Bk** and a distance **R2** from the rotation center line **5a** to the engaging portions **96** and **97** hold a relationship given by $R1 < R2$, and the distances **R1** and **R2** are different distances. The distance **R1** and the distance **R2** are the distances from the rotation center line **5a** to the connecting portion **95Bk** and the engaging portions **96** and **97** in a direction orthogonal to the rotation center line **5a**, and are substantially the rotation radii at the time of rotation of the head set cover **8**. The rotation radius of the engaging portions **96** and **97** is longer than that of the connecting portion **95Bk**.

The connecting member **9Cr** will be described. The basic structure of the connecting member **9Cr** is the same as the connecting member **9Bk**. The connecting member **9Cr** includes the main body **90** and the sealing member **91**. The main body **90** of the connecting member **9Cr** integrally includes flow-in portions **93C**, **93M**, and **93Y** provided at the front end portion, the pair of claw portions **92** provided on the left and right side portions, and connecting portions **95C**, **95M**, and **95Y** and the engaging portions **96** and **97** which are provided on the bottom surface. The tube **11** that forms the ink supply passage for cyan ink, the tube **11** that forms the ink supply passage for magenta ink, and the tube **11** that forms the ink supply passage for yellow ink are connected to the flow-in portions **93C**, **93M**, and **93Y**, respectively.

The connecting portions **95C**, **95M**, and **95Y** are members connected to the ink introduction portions **103C**, **103M**, and **103Y** of the printhead **10Cr**, respectively. The connecting portions **95C**, **95M**, and **95Y** according to this embodiment are tubular members, and the connecting ports **95a** to/from

which the ink introduction portions **103C**, **103M**, and **103Y** are inserted/removed are formed in these. The elastic member **95b** is provided on the inner circumferential surface of each of the connecting portions **95C**, **95M**, and **95Y**. The elastic members **95b** are sealing elastomers that seal the connecting ports **95a** and the ink introduction portions **103C**, **103M**, and **103Y** and are, for example, molded integrally with the connecting portions **95C**, **95M**, and **95Y**.

Groove-shaped relay passages **94C**, **94M**, and **94Y** through which the flow-in portions **93C**, **93M**, and **93Y** and the connecting portions **95C**, **95M**, and **95Y** communicate are formed on the upper surface of the main body **90** of the connecting member **9Cr**. The sealing member **91** is, for example, a sheet-shaped film welded to the upper surface of the main body **90**, and the relay passages **94C**, **94M**, and **94Y** are sealed by the sealing member **91**. When the connecting portion **95C** and the ink introduction portion **103C** are connected, the ink supply passage of the tube **11** and the ink introduction portion **103C** communicate, and cyan ink is supplied from the container **7C** to the printhead **10Cr** via the tube **11** and the relay passage **94C**. When the connecting portion **95M** and the ink introduction portion **103M** are connected, the ink supply passage of the tube **11** and the ink introduction portion **103M** communicate, and magenta ink is supplied from the container **7M** to the printhead **10Cr** via the tube **11** and the relay passage **94M**. When the connecting portion **95Y** and the ink introduction portion **103Y** are connected, the ink supply passage of the tube **11** and the ink introduction portion **103Y** communicate, and yellow ink is supplied from the container **7Y** to the printhead **10Cr** via the tube **11** and the relay passage **94Y**.

The pair of claw portions **92** of the connecting member **9Cr** are plate-shaped members and project from the main body **90** in the left-and-right direction (the direction of the axis **5a** of rotation) in a plan view of the connecting member **9Cr**. Like the connecting member **9Bk**, when the claw portions **92** are inserted into the opening portions **OP**, the connecting member **9Cr** is held by the head set cover **8** such that its posture can be changed. The engaging portions **96** and **97** correspond to the posture control portions **104** and **105** of the printhead **10Cr**, respectively. The engaging portion **96** is arranged to engage with the posture control portion **104**, and the engaging portion **97** is arranged to engage with the posture control portion **105**. The engaging portions **96** and **97** are tubular members. The engaging portion **96** has a cylindrical shape in correspondence with the shape of the posture control portion **104**, and the engaging portion **97** has an elliptic cylindrical shape in correspondence with the shape of the posture control portion **105**.

As shown in FIG. 6, distances **R3** and **R5** from the rotation center line **5a** to the connecting portion **95M** and the connecting portions **95C** and **95Y** and a distance **R4** from the rotation center line **5a** to the engaging portions **96** and **97** hold a relationship given by $R3 < R4 < R5$, and the distances **R3** to **R5** are different distances. The distances **R3** to **R5** are the distances from the rotation center line **5a** to the connecting portion **95M** and the connecting portions **95C** and **95Y** and the engaging portions **96** and **97** in a direction orthogonal to the rotation center line **5a**, and are substantially the rotation radii at the time of rotation of the head set cover **8**. The rotation radius of the engaging portions **96** and **97** is longer than that of the connecting portion **95M**. The rotation radius of the engaging portions **96** and **97** is shorter than that of the connecting portions **95C** and **95Y**.

Referring to FIG. 5, broken lines **D3** indicate the extending directions (axial directions) of the connecting portions **95Bk**, **95Y**, **95M**, and **95C** (to be referred to as connecting

portions **95** hereinafter generically or without distinction). The broken lines **D3** also indicate the connecting directions between the connecting portions **95Bk**, **95Y**, **95M**, and **95C** and the ink introduction portions **103Bk**, **103Y**, **103M**, and **103C**. Broken lines **D4** indicate the extending directions (axial directions) of the engaging portions **96** and **97**, and also indicate the connecting directions between the engaging portions **96** and **97** and the posture control portions **104** and **105**. In this embodiment, the directions **D3** and the directions **D4** are parallel.

<Positioning Portion>

FIGS. **7** and **8** will mainly be referred to. A positioning portion **14** is provided on the head set cover **8**. The positioning portion **14** is an engaging member biased by an elastic member **15** in a direction to press the printhead **10**. In this embodiment, the elastic member **15** is a coil spring. Two sets of positioning portions **14** and elastic members **15** are provided, and the sets correspond to the printheads **10Bk** and **10Cr**, respectively. When the head set cover **8** rotates from the open position to the closed position, the positioning portion **14** engages with the engaging portion **106** of the printhead **10**, and the printhead **10** is positioned at a predetermined position of the carriage **5**.

<Biasing Portion>

The printing apparatus **1** includes a biasing portion **13** that biases the connecting member **9** to a predetermined posture. In this embodiment, the biasing portion **13** is an elastic member provided for each of the connecting members **9Bk** and **9Cr** and is particularly a coil spring. The biasing portion **13** is arranged between the upper wall portion **8a** of the head set cover **8** and the connecting member **9**. The head set cover **8** is provided with a shaft-shaped guide portion **83** that supports the biasing portion **13** (FIGS. **7** and **8**). The biasing portion **13** contacts the sealing member **91** of the connecting member **9**. FIG. **9A** is a perspective view of the biasing portion **13**, FIG. **9B** is a plan view of the biasing portion **13**, and FIG. **9C** is a sectional view taken along a line B-B in FIG. **9B**.

In this embodiment, the biasing portion **13** is configured to come into direct contact with the sealing member **91**. Wire end portions **13a** and **13b** of the biasing portion **13** are bent inward in the axial direction. The wire end portions **13a** and **13b** are processed so as not to damage the sealing member **91** regardless of which end of the biasing portion **13** contacts the sealing member **91**.

The biasing point of the biasing portion **13** to the connecting member **9** is set to a point apart from the claw portions **92** in the direction orthogonal to the rotation center line **5a**, and in this embodiment, set to a point farther from the rotation center line **5a** than the claw portions **92** (a point on the side of the flow-in portion **93**).

FIG. **7** shows the initial posture of the connecting member **9Bk**, and FIG. **8** shows the connecting posture of the connecting member **9Bk**. The biasing portion **13** biases the connecting member **9Bk** to the initial posture. The initial posture is the posture of the connecting member **9Bk** in a state in which the connecting portion **95Bk** is not connected to the ink introduction portion **103Bk**, and the posture control portions **104** and **105** are apart from the engaging portions **96** and **97**. The connecting posture is the posture in a state in which the connecting portion **95Bk** is connected to the ink introduction portion **103Bk**. In the connecting posture, the direction **D1** and the direction **D3** match, and the direction **D2** and the direction **D4** match. It can be said that the connecting posture is a posture in which the connecting

portion **95Bk** is directed to the connecting direction to the ink introduction portion **103Bk** more than in the initial posture.

A line **L1** indicates a direction orthogonal to the direction **D3** (and the direction **D4**) in the initial posture, and a line **L2** indicates a direction orthogonal to the direction **D3** (and the direction **D4**) in the connecting posture. It can be said that the line **L1** and the line **L2** are the reference lines of the posture of the connecting member **9Bk** and correspond to the front-and-rear direction of the main body **90**. As for the posture of the connecting member **9Bk** with respect to the head set cover **8**, when the line **L2** is aligned in the front-and-rear direction of the head set cover **8**, the posture changes by an angle θ between the initial posture and the connecting posture. The biasing portion **13** need only bias the connecting member **9Bk** such that it tilts by the angle θ with respect to the head set cover **8**, and its elastic force can be relatively small. The connecting member **9Bk** has been described above, but the same can apply to the connecting member **9Cr**.

"3. Operation"

The behavior and the like of the connecting member **9** when the head set cover **8** is rotated from the open position to the closed position will be described with reference to FIGS. **10** to **13**. FIG. **10** is a sectional view taken along a line C-C in FIG. **3**, and is a view for explaining the operation, showing the rotation mode of the head set cover **8**. FIG. **11** is an enlarged view of a portion **P1** in FIG. **10**. FIG. **12** is a sectional view taken along a line D-D in FIG. **3**, and is a view for explaining the operation, showing the rotation mode of the head set cover **8**. FIG. **13** is an enlarged view of a portion **P2** in FIG. **12**. States **ST1** to **ST4** in FIGS. **10** to **13** represent the same operation timings.

The state **ST1** shows a state in which the head set cover **8** is at the open position, and the state **ST4** shows a state in which the head set cover **8** is at the closed position. At the open position, the connecting member **9** is in the initial posture. At the closed position, the connecting member **9** is in the connecting posture. As for the relationship between the ink introduction portion **103** and the connecting portion **95**, the closed position can be called a connecting position, and the open position can be called a connection releasing position. The states **ST2** and **ST3** are states halfway through rotation between the state **ST1** and the state **ST4**.

FIGS. **10** to **13** show a transition such as a posture change of the connecting member **9Cr**. Note that this also applies to a transition such as a posture change of the connecting member **9Bk**.

In the state **ST1**, the connecting member **9Cr** is in the initial posture. As indicated by the state **ST1** in FIG. **11**, the opening portion **OP** of the holding portion **81** is a pentagonal opening portion with linear edges **81a** to **81e**. The edge **81a** and the edge **81b** are parallel, and the edge **81c** and the edge **81d** are parallel. The edge **81e** is a tilted edge between the edge **81a** and the edge **81d**.

In the initial posture, the claw portion **92** of the connecting member **9Cr** is in contact with the edges **81a** to **81e** of the opening portion **OP** at a plurality of points. More specifically, the claw portion **92** is in contact with the edges of the opening portion **OP** at three points of the edges **81a**, **81b**, and **81e**. Since the opening portion **OP** is formed into not a rectangular shape but a pentagonal shape, the positioning accuracy of the connecting member **9Cr** can be improved while smoothly performing the posture change of the connecting member **9Cr**.

The operation of rotating the head set cover **8** from the open position to the closed position is performed by, for

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example, the user. When the head set cover **8** starts rotating counterclockwise in FIGS. **10** to **13**, the state ST2 is obtained. At this stage, as indicated by the state ST2 in FIGS. **11** and **13**, the posture control portion **105** and the corresponding engaging portion **97** start engaging. More specifically, the shaft portion **105a** starts entering the hole **97a**. This also applies to the shaft portion **104a** of the posture control portion **104** and the corresponding engaging portion **96**. On the other hand, at this stage, the ink introduction portion **103** and the connecting portion **95** are apart. That is, in the process of rotation of the head set cover **8** from the open position to the closed position, the posture control portions **104** and **105** and the engaging portions **96** and **97** start contacting (or engaging) before the ink introduction portion **103** and the connecting portion **95**. Also, although not illustrated, at the stage of the state ST2, the positioning portion **14** engages with the engaging portion **106** of the printhead **10**, and the printhead **10** starts to be positioned at a predetermined position of the carriage **5**.

When the rotation of the head set cover **8** progresses, the state ST3 is obtained. At this stage, as indicated by the state ST3 in FIGS. **11** and **13**, the engagement between the posture control portion **105** and the corresponding engaging portion **97** progresses. This also applies to the posture control portion **104** and the corresponding engaging portion **96**. When the shaft portion **105a** enters the hole **97a**, axis adjustment for positioning both on the same axis acts. That is, axis adjustment for aligning the direction D2 and the direction D4 acts. Accordingly, the connecting member **9Cr** changes its posture against the biasing of the biasing portion **13**, and the claw portion **92** starts separating from the edges **81a** and **81e**. The posture of the connecting member **9Cr** becomes close to the connecting posture, the direction D1 and the direction D3 start to be aligned, and in this state, the ink introduction portion **103** and the connecting portion **95** start to be connected. Also, by the engagement between the positioning portion **14** and the engaging portion **106** of the printhead **10**, the printhead **10** starts to be positioned at a predetermined position of the carriage **5**.

When the rotation of the head set cover **8** progresses, the state ST4 is obtained. At this stage, the posture of the connecting member **9Cr** reaches the connecting posture. Thus, in the rotation process of the head set cover **8**, the posture of the connecting member **9Cr** can automatically be changed. As indicated by the state ST4 in FIGS. **11** and **13**, the engagement between the posture control portion **105** and the corresponding engaging portion **97** is completed, and the connection between the ink introduction portion **103** and the connecting portion **95** is completed. The claw portion **92** is in contact with the edge **81b** with a surface. The lock portion **82** is locked with the lock portion **52**, and the head set cover **8** is maintained at the closed position. The printhead **10** is fixed to the carriage **5**.

As described above, in this embodiment, by the rotation of the head set cover **8** from the open position to the closed position, fixing of the printhead **10** and connection between the ink introduction portion **103** and the connecting portion **95** can simultaneously be performed, and ink can be supplied from the container **7** to the printhead **10** immediately. When the lock between the lock portion **82** and the lock portion **52** is released, the head set cover **8** automatically rotates to the open position by the biasing of the elastic member **12**. At this time, the connecting member **9** returns to the initial posture by an operation reverse to the states ST1 to ST4, and the connection between the ink introduction portion **103** and the connecting portion **95** is released. By only releasing the lock between the lock portion **82** and the

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lock portion **52**, the user can detach the printhead **10** from the carriage **5** and exchange it.

In this embodiment, the following advantage can be obtained by providing the biasing portion **13** and the holding portions **81** as a mechanism for changing the posture of the connecting member **9**. That is, immediately before the ink introduction portion **103** and the connecting portion **95** are connected, the connecting member **9** can be changed to a posture suitable for the connection between these along with the rotation of the head set cover **8**, and the connection accuracy can be improved. In particular, it is possible to reduce the possibility that the ink introduction portion **103** and the connecting portion **95** start to be obliquely connected to each other, and a connection failure occurs.

Since the connecting member **9** is always biased by the biasing portion **13** and pressed against the edges of the opening portion OP, the posture of the connecting member **9** is stable. During the rotation of the head set cover **8**, the posture of the connecting member **9** can be prevented from changing to an unintended posture. From this viewpoint as well, it is possible to reduce the possibility of a connection failure between the ink introduction portion **103** and the connecting portion **95** and improve the connection accuracy.

The posture change of the connecting member **9** does not restrict the rotation of the head set cover **8**, and the head set cover **8** and the connecting member **9** can smoothly be displaced toward each other.

Also, by the configuration according to this embodiment, the connection accuracy between the ink introduction portion **103** and the connecting portion **95** can be improved while reducing the size of the head set cover **8**. This point will be described with reference to FIGS. **14A** and **14B**. FIG. **14A** shows, as a comparative example, an example of a configuration without the mechanism for changing the posture of the connecting member **9**. Along with the rotation of the head set cover **8**, the connecting portion **95** moves on an arc with a rotation radius R from the rotation center line **5a**.

If a connecting direction DO between the ink introduction portion **103** and the connecting portion **95** is substantially orthogonal to a line Dr that is orthogonal to the rotation center line **5a**, connection between the ink introduction portion **103** and the connecting portion **95** is smoothly done if the rotation radius R is long. This is because the moving direction of the connecting portion **95** is close to the connecting direction DO near the ink introduction portion **103**. However, if the rotation radius R is short, the angle of the moving direction of the connecting portion **95** crossing the connecting direction DO near the ink introduction portion **103** becomes large. Hence, a connection failure readily occurs. To improve the connection accuracy, the rotation radius R needs to be long, and this makes the apparatus bulky.

On the other hand, in this embodiment, as shown in FIG. **14B**, the posture of the connecting member **9** changes during the process of the rotation of the head set cover **8**, and the connecting portion **95** can be connected to the ink introduction portion **103** not by a circular motion but in a state close to a linear motion. Hence, even if the rotation radius R of the connecting portion **95** is short, the connection accuracy can be improved, and the size of the apparatus can be reduced.

Second Embodiment

In the first embodiment, the biasing portion **13** is arranged between the head set cover **8** and the connecting member **9**. However, the disposition point of a biasing portion **13** is not

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limited to this. For example, the biasing portion 13 may be arranged between a connecting member 9 and a printhead 10.

Also, in the first embodiment, a coil spring is used as the biasing portion 13. However, another component can be employed as the biasing portion 13. For example, rubber or a resin may be used.

Also, as the biasing portion 13, a tube 11 with flexibility may be used. FIG. 15 shows an example. In the example shown in FIG. 15, the tube 11 that forms an ink supply passage is used as a biasing portion 13', and the biasing portion 13 formed by a coil spring in the first embodiment is not provided. In a holding portion 84, the tube 11 is fixed to a head set cover 8 at its middle portion. Using the flexibility of the tube 11, the connecting member 9 can be biased to the initial posture, as indicated by an arrow D11. Note that three tubes 11 are used as the biasing portions 13' for a connecting member 9Cr, and one tube 11 is used as the biasing portion 13' for a connecting member 9Bk. The material, diameter, thickness, hardness, and length (extra length) from the holding portion 84 may be changed between the three tubes 11 connected to the connecting member 9Cr and the one tube 11 connected to the connecting member 9Bk. For example, if the biasing force is insufficient for the connecting member 9Bk, as the tube 11 for black ink, a tube having a larger thickness (stronger elastic force) or diameter than the tubes 11 for the remaining inks may be used. According to this embodiment, the number of parts can be decreased.

Also, the biasing portion 13 may use not the elastic force but gravity. For example, a sinker fixed to the connecting member 9 may be used. The sinker may be a member different from the connecting member 9, or may be a portion which is integrated with the connecting member 9 and whose weight is increased by making the portion thicker than the remaining portions.

Third Embodiment

A claw portion 92 and an opening portion OP can employ various structures.

FIG. 16A shows an example. In the example shown in FIG. 16A, the claw portion 92 is a member having a semicircular section. The opening portion OP has a rectangular shape. In FIG. 16A, a state ST11 indicates a state in which a connecting member 9 is in an initial posture, a state ST13 indicates a state in which the connecting member 9 is in a connecting posture, and a state ST12 indicates a state in which the connecting member 9 is in a posture between the initial posture and the connecting posture. In the initial posture indicated by the state ST11, the plane portion of the claw portion 92 is in contact with an edge of the opening portion OP with a surface. Since the claw portion 92 and the edge of the opening portion OP are in surface contact, the posture of the connecting member 9 is stable. In the state ST12, the claw portion 92 does not contact the edge of the opening portion OP. In the state ST13, the arc circumferential surface portion of the claw portion 92 is in contact with an edge of the opening portion OP at one point.

FIG. 16B shows another example. In the example shown in FIG. 16B, the claw portion 92 is a member having a circular section. The opening portion OP has an elliptic shape. In FIG. 16B, a state ST21 indicates a state in which the connecting member 9 is in the initial posture, a state ST23 indicates a state in which the connecting member 9 is in a connecting posture, and a state ST22 indicates a state in which the connecting member 9 is in a posture between the

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initial posture and the connecting posture. In the initial posture indicated by the state ST21, the circumferential surface portion of the claw portion 92 is in contact with an edge of the opening portion OP at one point. In the state ST22, the claw portion 92 does not contact the edge of the opening portion OP. In the state ST23, the circumferential surface portion of the claw portion 92 is in contact with an edge of the opening portion OP at one point.

In the configuration shown in FIG. 16B, regulation to the posture of the connecting member 9 may be weak. Hence, an example of a configuration shown in FIG. 17 can be used together. In the example of the configuration shown in FIG. 17, a head set cover 8 is provided with a contact portion 85, and a connecting member 9Bk is provided with a contact portion 98 that comes into contact with the contact portion 85. The contact surface of the contact portion 85 is formed as a tilting surface, and the contact portion 98 contacts the tilting surface. Note that the head set cover 8 is provided with the contact portion 85 for a connecting member 9Cr, and the connecting member 9Cr is provided with the corresponding contact portion 98.

FIG. 17 shows a state in which the head set cover 8 is located at the open position, and the connecting member 9Bk is in an initial posture. The posture of the connecting member 9Bk can be regulated by the angle of the tilting surface of the contact portion 85 and the shape of the contact portion 98.

Fourth Embodiment

In the above-described embodiments, an example of a configuration in which a plurality of types of posture control portions 104 and 105 with different sectional shapes and corresponding engaging portions 96 and 97 are provided has been described. However, there may be only one type of posture control portions and corresponding engaging portions. Also, in the above-described embodiments, for one set of a printhead 10 and a connecting member 9, a plurality of sets of posture control portions and corresponding engaging portions are provided. However, only one set may be provided.

In the above-described embodiments, the posture control portions 104 and 105 are provided on the printhead 10. However, these may be provided on another point. For example, the posture control portions may be provided on a carriage 5, and the positions of the engaging portions 96 and 97 are designed in accordance with the disposition points of the posture control portions.

In the above-described embodiments, as the posture control portions 104 and 105, structures including a shaft portion 104a having a columnar shape (circular section) and a shaft portion 105a having an elliptic columnar shape (elliptic section) have been described. However, the structures of the posture control portions are not limited to these, and various shapes can be employed.

The number of ink introduction portions 103, the distance between each ink introduction portion 103 and the rotation center line 5a, the types and the number of posture control portions 104 and 105, and the distance between each of the posture control portions 104 and 105 and the rotation center line 5a can appropriately be designed. The initial posture and the connecting posture of the connecting member 9 are also appropriately designed in accordance with the configuration of the apparatus.

Other Embodiments

Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads

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out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) 5 and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium 10 to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), 25 digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary 30 embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-073671, filed Apr. 27, 2022, which is hereby incorporated by reference herein in its entirety. 35

What is claimed is:

1. A printing apparatus comprising:

a support member configured to support a printhead that performs printing by discharging ink to a print medium; 40

a connecting member including a connecting portion connected to an ink introduction portion of the printhead and through which the ink introduction portion communicates with an ink supply passage via the connecting portion; 45

a rotation member configured to hold the connecting member such that a posture of the connecting member is changeable, and configured to be supported by the support member to be rotatable between a connecting position at which the ink introduction portion and the connecting portion are connected and a connection releasing position at which the ink introduction portion and the connecting portion are not connected; and 50

a biasing portion configured to bias the connecting member to a first posture with respect to the rotation member, 55

wherein the connecting member includes an engaging portion configured to engage with a posture control portion provided at a predetermined point, and during a process of rotation of the rotation member from the connection releasing position to the connecting position, the engaging portion engages with the posture control portion, and the connecting member changes from the first posture to a second posture against the biasing of the biasing portion. 65

2. The apparatus according to claim 1, wherein the posture control portion is provided on the printhead.

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3. The apparatus according to claim 1, wherein the second posture is a posture in which the connecting portion is directed in a connecting direction further toward the ink introduction portion than in the first posture.

4. The apparatus according to claim 1, wherein the rotation member includes at least one opening portion configured to hold the connecting member, the connecting member includes a claw portion configured to be inserted into the opening portion, and by the biasing of the biasing portion, the claw portion contacts an edge of the opening portion, and the connecting member is maintained in the first posture.

5. The apparatus according to claim 4, wherein in the first posture, the claw portion contacts the edge of the opening portion at a plurality of points.

6. The apparatus according to claim 4, wherein in the first posture, the claw portion contacts the edge of the opening portion with a surface.

7. The apparatus according to claim 4, wherein in the second posture, the claw portion contacts the edge of the opening portion.

8. The apparatus according to claim 4, wherein the at least one opening portion opens in a rotation axis direction of the rotation member, and includes two opening portions which are provided apart in the rotation axis direction, and

the claw portion projects in the rotation axis direction, and one claw portion is provided in correspondence with each opening portion.

9. The apparatus according to claim 4, wherein one of the posture control portion and the engaging portion includes a shaft portion, and the other includes a tube portion to/from which the shaft portion is inserted/removed.

10. The apparatus according to claim 9, wherein an axial direction of the shaft portion and the tube portion is parallel to a connecting direction between the connecting portion and the ink introduction portion.

11. The apparatus according to claim 9, wherein a distance between the engaging portion and a rotation center of the rotation member and a distance between the connecting portion and the rotation center of the rotation member are different, and

a distance between the posture control portion and the rotation center of the rotation member and a distance between the ink introduction portion and the rotation center of the rotation member are different.

12. The apparatus according to claim 1, wherein the rotation member includes:

an opening portion configured to hold the connecting member; and

a contact portion that the connecting member contacts, the connecting member includes a claw portion to be inserted into the opening portion, and by the biasing of the biasing portion, the connecting member contacts the contact portion, and the connecting member is maintained in the first posture.

13. The apparatus according to claim 1, wherein the rotation member includes a cover portion configured to cover the printhead, and at the connecting position, the connecting member is located between the cover portion and the printhead.

14. The apparatus according to claim 13, wherein the biasing portion is an elastic member arranged between the cover portion and the connecting member.

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15. The apparatus according to claim 13, wherein the biasing portion is a flexible tube connected to the connecting member and forming the ink supply passage.

16. The apparatus according to claim 1, wherein the support member is a carriage configured to move in a widthwise direction of the print medium.

17. The apparatus according to claim 1, wherein the printhead is detachably supported by the support member, and

the rotation member includes a positioning portion configured to position the printhead with respect to the support member at the connecting position.

18. The apparatus according to claim 1, wherein the support member includes a lock portion configured to maintain the rotation member at the connecting position.

19. A printing apparatus comprising:

a printhead configured to perform printing by discharging ink to a print medium;

a support member configured to support the printhead;

a connecting member including a connecting portion connected to an ink introduction portion of the printhead and through which the ink introduction portion communicates with an ink supply passage via the connecting portion;

a rotation member configured to hold the connecting member such that a posture of the connecting member is changeable, and configured to be supported by the support member to be rotatable between a connecting position at which the ink introduction portion and the connecting portion are connected and a connection releasing position at which the ink introduction portion and the connecting portion are not connected; and

a biasing portion configured to bias the connecting member to a first posture with respect to the rotation member,

wherein the connecting member includes an engaging portion configured to engage with a posture control portion provided at a predetermined point, and

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during a process of rotation of the rotation member from the connection releasing position to the connecting position, the engaging portion engages with the posture control portion, and the connecting member changes from the first posture to a second posture against the biasing of the biasing portion.

20. A printhead exchangeably mounted in a printing apparatus, wherein

the printhead comprises an ink introduction portion, and the printing apparatus includes:

a support member configured to support the printhead;

a connecting member including a connecting portion connected to an ink introduction portion of the printhead and through which the ink introduction portion communicates with an ink supply passage via the connecting portion;

a rotation member configured to hold the connecting member such that a posture of the connecting member is changeable, and configured to be supported by the support member to be rotatable between a connecting position at which the ink introduction portion and the connecting portion are connected and a connection releasing position at which the ink introduction portion and the connecting portion are not connected; and

a biasing portion configured to bias the connecting member to a first posture with respect to the rotation member,

wherein the printhead includes a posture control portion, the connecting member includes an engaging portion configured to engage with the posture control portion, and

during a process of rotation of the rotation member from the connection releasing position to the connecting position, the engaging portion engages with the posture control portion, and the connecting member changes from the first posture to a second posture against the biasing of the biasing portion.

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